Supplementary Information

The Star of Bethlehem belongs to the guiding-star legendary narrative tradition: evidence from orbital mechanics and Greek corpus linguistics

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S1 Orbital mechanics: detailed methods

S1.1 Reference epoch and coordinate system

All calculations use the International Celestial Reference System (ICRS, J2000.0) as the inertial frame. The reference observation window is Julian Day 1 719 755.9 (TDB), corresponding to ≈ 08:00 local solar time at Bethlehem on a date in 5 BCE. The UT1–UTC offset appropriate for 5 BCE (ΔT) is 10,572 s, derived from the long-term IERS extrapolation of [1]. Observer location: Bethlehem, 31.70° N, 35.20° E, altitude 765 m.

S1.2 East–North–Up rotation matrices

Transformation from ICRS to local horizontal (azimuth–altitude) coordinates was performed via pre-computed East–North–Up (ENU) rotation matrices. At each time step the matrix was constructed by converting AltAz basis vectors to ICRS using `astropy`'s `AltAz`→`SkyCoord.icrs` transformation, which accounts for Earth's precession, nutation, and Greenwich Apparent Sidereal Time. The basis vectors were computed as follows:

- East: pointing toward (Az = 90°, Alt = 0.001°)
- Zenith: pointing toward (Az = 0°, Alt = 90°)
- North: $\hat{N} = \hat{Z} \times \hat{E}$

This approach avoids the 25.7° systematic azimuth error that arises from applying J2000.0 hour-angle formulae directly to 5 BCE observations without precession correction. The matrices were validated by recovering the known azimuth of Matney's proposed comet[2] (193.7°) to within 0.01° at the reference epoch.

S1.3 Heliocentric orbit propagation

Positions of solar-orbiting bodies were computed from osculating orbital elements using the appropriate Kepler equation for each orbit type.

Parabolic orbits ($e = 1$). Barker's equation gives the true anomaly ν directly:

$$D = \sqrt[3]{\frac{W}{2} + \sqrt{\frac{W^2}{4} + 1}} + \sqrt[3]{\frac{W}{2} - \sqrt{\frac{W^2}{4} + 1}}, \qquad \nu = 2 \arctan D, \tag{1}$$

where $W = 3\sqrt{\mu_\odot / 2q^3}\, \Delta t$, $\mu_\odot = k^2 = 2.959 \times 10^{-4}$ AU^3 day^{-2}, and $k = 0.01720209895$ is the Gaussian gravitational constant.

Elliptic orbits ($0 \le e < 1$)**.** Mean anomaly $M = n(t - T)$ where $n = \sqrt{\mu_\odot / a^3}$. Eccentric anomaly E is solved from Kepler's equation $M = E - e \sin E$ by Newton–Raphson iteration to a tolerance of 10^{-12} rad. True anomaly:

$$\nu = 2 \arctan\left(\sqrt{\frac{1+e}{1-e}} \tan \frac{E}{2} \right). \tag{2}$$

Hyperbolic orbits ($e > 1$)**.** The hyperbolic Kepler equation $M = e \sinh H - H$ is solved for the hyperbolic anomaly H by Newton–Raphson. True anomaly:

$$\nu = 2 \arctan\left(\sqrt{\frac{e+1}{e-1}} \tanh \frac{H}{2} \right), \qquad |\nu| < \arccos(-1/e). \tag{3}$$

Earth positions were drawn from the JPL DE430 ephemeris. For computational efficiency, heliocentric positions were vectorised over the longitude of ascending node Ω and argument of perihelion ω using pre-computed perifocal unit vectors $\hat{P}(\Omega, \omega, i)$ and $\hat{Q}(\Omega, \omega, i)$ with `NumPy` `einsum` operations, enabling evaluation of $72 \times 72 = 5{,}184$ orientations per loop iteration at the 5° grid resolution used in the fine survey.

S1.4 Geocentric orbit propagation

Geocentric configurations used the same Kepler equation solver with $GM_\oplus = 3.986 \times 10^5\ \mathrm{km^3\,s^{-2}}$. The observer's position in the Geocentric Celestial Reference System (GCRS) was computed at each time step from `EarthLocation.get_gcrs()` in `astropy`, accounting for Earth's rotation. The direction from observer to satellite was converted to AltAz via the ENU rotation matrix. Configurations with satellite altitude below 5° at any guidance time step were excluded from the survey totals.

S1.5 Guidance and stopping criteria

Site coordinates. Jerusalem Old City center: 31°46′ N, 35°14′ E. Bethlehem (Church of the Nativity): 31°42′ N, 35°12′ E. Straight-line bearing: 203°. The ridge road sweeps from ≈ 190° to ≈ 210° along the route.

Angular-velocity threshold. We adopt $\omega_{\text{thresh}} = 2^\circ\,\mathrm{h}^{-1}$ for both criteria. *Physical basis:* naked-eye angular resolution is ≈ 1′; motion relative to background stars is therefore detectable at any rate above $\approx 0.03^\circ\,\mathrm{h}^{-1}$ over a 30-minute observation window. For comparison, the Moon moves $\approx 0.5^\circ\,\mathrm{h}^{-1}$ relative to background stars—a rate routinely noted by ancient Babylonian observers. Our threshold of $2^\circ\,\mathrm{h}^{-1}$ is 4× the lunar rate, detectable within minutes. The choice is deliberately generous: any stricter threshold only strengthens the impossibility conclusion.

Guidance criterion (three components).

1. *Visibility:* object altitude > 5° above the horizon throughout the guidance window (08:00–10:00 local time).
2. *Azimuth:* object azimuth remains within the road corridor $[190^\circ, 210^\circ] \pm 5^\circ$ at all 9 steps (15-min spacing). Guidance score = maximum degrees outside $[190^\circ, 210^\circ]$; passes if $< 5^\circ$.
3. *Motion (προάγω):* minimum apparent angular velocity $> \omega_{\text{thresh}} = 2^\circ\,\mathrm{h}^{-1}$ throughout the guidance window. A stationary object at the right azimuth does not "go before" travellers; this criterion excludes such configurations.

Stopping criterion (ἵστημι). Maximum apparent angular velocity $< \omega_{\text{thresh}} = 2^\circ\,\text{h}^{-1}$ during the stopping window (10:15–12:00, 8 steps). No altitude constraint is imposed: scholarly astronomical interpretations of "stood over" (ἐστάθη ἐπάνω) range from horizon to zenith.

Sensitivity to threshold choice. Table S1 shows survey results as ω_{thresh} is varied over two orders of magnitude. Zero configurations satisfy all three criteria at every tested value.

Table S1: Sensitivity of the zero result to angular-velocity threshold ω_{thresh} (heliocentric elliptic/parabolic survey, 5 BCE reference epoch).

ω_{thresh} (°/h)	N_az	N_motion	N_stop	N_all
0.25	0	201	49,196	0
0.50	0	25	49,381	0
1.00	0	16	49,389	0
2.00	0	0	49,406	0
5.00	0	0	49,406	0
10.00	0	0	49,406	0

N_az = configurations passing azimuth criterion;
N_motion = configurations with guidance motion $> \omega_{\text{thresh}}$;
N_stop = configurations with stopping score $< \omega_{\text{thresh}}$;
N_all = configurations passing all three criteria simultaneously.

The binding constraint is the azimuth criterion (N_az = 0 at every threshold): no Keplerian orbit keeps the object within the road corridor throughout the 2-hour guidance window. The motion constraint further tightens the result at lower thresholds by excluding near-stationary objects that coincidentally lie near 200° azimuth.

S1.6 Orbital parameter survey grids

Table S2: Orbital parameter grid for the heliocentric survey.

Parameter	Value
Perihelion distance q (AU)	0.02, 0.03, 0.04, 0.05, 0.07, 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, 0.40, 0.50, 0.6
Eccentricity e	0, 0.1, 0.2, 0.3, 0.5, 0.7, 0.8, 0.9, 0.95, 0.99, 1.0, 1.1, 1.5, 2.0, 5.0 (15 value
Inclination i	0°–180° in 5° steps plus $1^\circ, 1.4^\circ, 2^\circ, 3^\circ, 4^\circ$ (42 total values)
Perihelion epoch offset ΔT (days)	−60, −40, −20, −10, −5, +5, +10, +20, +60 (9 values)
Ω and ω	0°–355° in 5° steps (72 values each)
Outer combinations	$18 \times 15 \times 42 \times 9 = 102{,}060$
Orientations per combination	$72 \times 72 = 5{,}184$
Total configurations sampled	529,079,040
Total visible	202,372,076
Satisfying all three criteria	**0**

Note on inclination range. Inclination i is defined as the dihedral angle between the orbital plane and the ecliptic reference plane; by construction $i \in [0^\circ, 180^\circ]$. An inclination $i < 0^\circ$ is physically equivalent to $(|i|, \Omega + 180^\circ)$, and $i > 180^\circ$ is equivalent to $(360^\circ - i, \Omega + 180^\circ)$. Since Ω is swept over all 360° in both surveys, no physically distinct orbital plane is omitted by restricting i to this range.

Table S3: Orbital parameter grids for the geocentric survey.

Parameter	Values
Semi-major axis a	7,000 to 500,000 km (LEO to beyond lunar)
Eccentricity e	0, 0.2, 0.5, 0.7, 0.8, 0.9, 0.95, 0.99 (8 values)
Inclination i	$0°, 45°, 90°, 135°, 180°$ (5 values)
Argument of perigee ω	$0°, 90°, 180°, 270°$ (4 values)
RAAN Ω	$0°, 90°, 180°, 270°$ (4 values)
Initial mean anomaly M_0	0°–315° in 45° steps (8 values)
Total configurations	$10 \times 8 \times 5 \times 4 \times 4 \times 8 = 51{,}200$ (28,800 with perigee altitude > 100 km)
Timestep	5 min over 12-hour nighttime window
Stopping criterion used	$\omega_{\text{stop}} < 0.20° \, \text{h}^{-1}$ (strict); $< 0.50° \, \text{h}^{-1}$ (loose)
All-criteria pass count	**0** (strict); 8 (loose, all transit artifact; see § S1.10)

S1.7 Monte Carlo refinement sweeps

The 5° angular-grid spacing in the heliocentric survey leaves gaps of up to ∼3.5° between neighboring grid points. To verify that the null result is not an artefact of this resolution, two Monte Carlo sweeps in continuous parameter space were performed using the script `sob_mc_refined.py` (same Kepler solvers, ENU matrices, and evaluation criteria as `sob_proof_fine.py`).

Sweep 1: neighborhood of best-fit grid orbit. The best azimuth score in the exhaustive grid survey (0.015° inside the road corridor) belongs to the orbit with $e = 1.1$, $q = 0.03$ AU, $i = 10°$. Sweep 1 sampled 1,500,000 configurations drawn uniformly from

$$q \in [0.005, 0.15] \text{ AU}, \quad e \in [0.7, 2.5], \quad i \in [0°, 40°], \quad \Omega, \omega \in [0°, 360°], \quad \Delta T \in [-60, +60] \text{ days},$$

spanning a generous volume around the grid orbit. Of 601,150 visible configurations, zero satisfied all three criteria simultaneously. The orbit with the best azimuth score (0.167° from corridor center) produced a guidance-phase angular velocity of only $0.006° \, \text{h}^{-1}$—a factor of ≈330 below the required $2° \, \text{h}^{-1}$. The orbit with the highest guidance motion anywhere in the sweep ($93.5° \, \text{h}^{-1}$) had an azimuth score of 178°—far outside the road corridor. This confirms the physical trade-off documented in the grid survey: objects moving fast enough to guide are never pointed in the right direction, and objects pointed in the right direction are nearly stationary.

Sweep 2: neighborhood of Matney's proposed 5 BCE comet. Matney (2025) proposed a sungrazing comet with $q = 0.0407$ AU, $e = 1.0$, $i = 1.43°$, $\omega = 335.58°$, $\Omega = 75.81°$, $T_{\text{peri}} =$ JD 1,719,785.565 as a candidate Star of Bethlehem. Sweep 2 sampled 750,000 configurations uniformly from

$$q \in [0.01, 0.16] \text{ AU}, \quad e \in [0.7, 1.5], \quad i \in [0°, 8°], \quad \Omega \in [26°, 126°], \quad \omega \in [286°, 386°], \quad \Delta T \in [-10, +70] \text{ days},$$

bracketing Matney's parameters with generous margins in every dimension. Of 294,672 visible configurations, zero satisfied all three criteria. The best azimuth score was 9.49°—outside the ±5° corridor threshold—and the guidance motion at that orbit was $0.006° \, \text{h}^{-1}$. The highest guidance motion found anywhere in the sweep ($0.84° \, \text{h}^{-1}$) is itself below the $2° \, \text{h}^{-1}$ threshold and occurs at an azimuth 141° outside the corridor. Fine-tuning around Matney's comet does not produce a viable candidate.

Joint conclusion. Across both sweeps, 2,250,000 configurations were sampled (895,822 visible) with zero satisfying all criteria. The null result of the exhaustive grid survey is confirmed in continuous parameter space: the impossibility is structural, not a consequence of grid spacing.

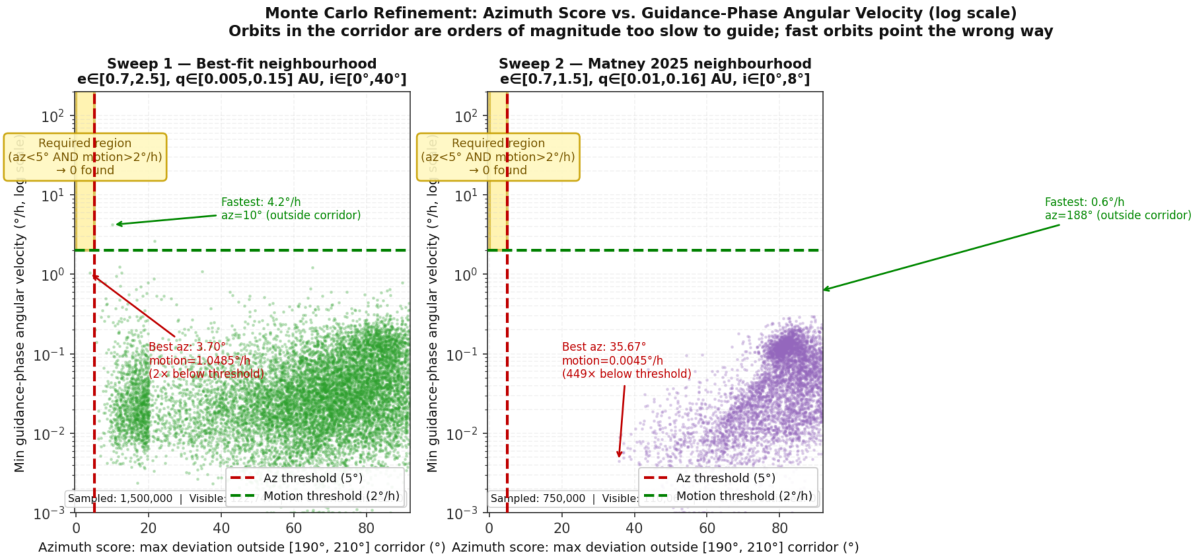

Figure S1: **Supplementary Figure S1: Monte Carlo refinement sweeps — azimuth score vs. guidance-phase angular velocity (log scale).** Each point is one visible orbital configuration from the two continuous-parameter sweeps described in §S1.7. **Left (Sweep 1):** 1.5 million configurations in the neighborhood of the best-fit grid orbit ($e \in [0.7, 2.5]$, $q \in [0.005, 0.15]$ AU, $i \in [0°, 40°]$, all Ω/ω, $\Delta T \in [-60, +60]$ days); 601,150 visible, 0 satisfying all criteria. **Right (Sweep 2):** 750,000 configurations in the neighborhood of Matney's (2025) proposed 5 BCE comet ($e \in [0.7, 1.5]$, $q \in [0.01, 0.16]$ AU, $i \in [0°, 8°]$, $\Omega \in [26°, 126°]$, $\omega \in [286°, 386°]$); 294,672 visible, 0 satisfying all criteria. In both panels the gold shaded region marks the zone where both the azimuth criterion ($< 5°$) and guidance-motion criterion ($> 2°\ \mathrm{h}^{-1}$) are simultaneously satisfied. The log scale reveals the structural anti-correlation: objects in the road corridor are orders of magnitude too slow to guide ($\sim 0.006°\ \mathrm{h}^{-1}$, a factor of ~ 330 below threshold in Sweep 1), while the fastest objects point in entirely the wrong direction.

Table S4: Monte Carlo refinement sweep results.

Sweep	Sampled	Visible	All OK	Best az score	Best motion
Best-fit neighborhood	1,500,000	601,150	0	0.167°	93.5° h^{-1} (az: 178° off)
Matney neighborhood	750,000	294,672	0	9.49° (outside)	0.84° h^{-1} (below thresh.)
Combined	2,250,000	895,822	**0**	—	—

Parameters sampled uniformly (seed 42). "All OK" = visible *and* azimuth within ±5° corridor *and* guidance motion > 2° h^{-1} *and* stopping motion < 2° h^{-1}. "Best az score" = minimum azimuth deviation from corridor center; "Best motion" = maximum guidance-phase angular velocity found anywhere in the sweep (with parenthetical showing why it still fails).

S1.8 Analytical derivation of the guidance–stopping correlation

Fig. 3 of the main text shows that every sampled Keplerian orbit falls near the diagonal $\omega_{\text{stop}} = \omega_{\text{guide}}$, regardless of orbit type (elliptic, parabolic, hyperbolic) or eccentricity. The caption notes that "Keplerian orbits cannot decelerate abruptly over four hours"; we derive this quantitatively here.

Apparent angular velocity in closed form. Let $\mathbf{d}(t) = \mathbf{r}_{\text{body}}(t) - \mathbf{r}_{\text{E}}(t)$ be the geocentric displacement vector, with $d = |\mathbf{d}|$ and unit vector $\hat{\mathbf{d}} = \mathbf{d}/d$. Let $\mathbf{V}(t) = \dot{\mathbf{r}}_{\text{body}} - \dot{\mathbf{r}}_{\text{E}}$ be the geocentric velocity. Decompose $\mathbf{V}$ into radial and transverse components, $\mathbf{V} = V_r\,\hat{\mathbf{d}} + \mathbf{V}_\perp$, with $V_r = \mathbf{V}\cdot\hat{\mathbf{d}}$ and $\mathbf{V}_\perp = \mathbf{V} - V_r\,\hat{\mathbf{d}}$. Then

$$\frac{\mathrm{d}\hat{\mathbf{d}}}{\mathrm{d}t} = \frac{\mathbf{V}}{d} - \frac{V_r}{d}\,\hat{\mathbf{d}} = \frac{\mathbf{V}_\perp}{d}, \tag{4}$$

so the apparent angular velocity is, exactly,

$$\omega_{\text{app}}(t) = \left|\frac{\mathrm{d}\hat{\mathbf{d}}}{\mathrm{d}t}\right| = \frac{V_\perp(t)}{d(t)}, \tag{5}$$

where $V_\perp = |\mathbf{V}_\perp|$.

Fractional rate of change. Differentiating (5) logarithmically,

$$\frac{1}{\omega_{\text{app}}}\frac{\mathrm{d}\omega_{\text{app}}}{\mathrm{d}t} = \frac{\dot{V}_\perp}{V_\perp} - \frac{\dot{d}}{d} = \frac{\dot{V}_\perp}{V_\perp} - \frac{V_r}{d}. \tag{6}$$

We bound each term separately.

Term 1: change in transverse velocity. The only force acting on the body (in the inertial frame, ignoring planetary perturbations) is solar gravity $\mathbf{a} = -\mu_\odot\,\mathbf{r}_{\text{body}}/r^3$. Its component perpendicular to the geocentric line of sight has magnitude at most $|\mathbf{a}| = \mu_\odot/r^2$ (for the body) plus $|\mathbf{a}_E| = \mu_\odot/r_E^2$ (for Earth, which shifts the reference frame). The transverse geocentric velocity satisfies $V_\perp \geq v_c/2$ over most of the orbit (where $v_c = \sqrt{\mu_\odot/r}$ is the local circular speed), so

$$\frac{|\dot{V}_\perp|}{V_\perp} \lesssim \frac{2\mu_\odot r^{-2}}{v_c} = \frac{2\mu_\odot}{r^2\sqrt{\mu_\odot/r}} = \frac{2\sqrt{\mu_\odot}}{r^{3/2}} = 2n_c, \tag{7}$$

where $n_c = \sqrt{\mu_\odot/r^3}$ is the local circular mean motion. For bound orbits $n_c = n\,(a/r)^{3/2}$ (with $n = 2\pi/T$), and for parabolic/hyperbolic orbits a similar expression holds with r replaced by the current heliocentric distance. Adding the Earth-acceleration term at most doubles the bound when $r \sim r_E$, giving $|\dot{V}_\perp|/V_\perp \lesssim 4n_c$.

Term 2: change in geocentric distance. $|\dot{d}| = |V_r| \le V_{\rm geo} \equiv |\mathbf{V}|$, so

$$\left|\frac{V_r}{d}\right| \le \frac{V_{\rm geo}}{d}. \tag{8}$$

For any heliocentric body with geocentric distance $d \ge d_{\min}$, $V_{\rm geo} \le V_{\rm body} + V_E \lesssim \sqrt{2\mu_\odot/r} + V_E$ (escape speed plus Earth's orbital speed $V_E \approx 29.8\,\mathrm{km\,s^{-1}}$).

Integrated bound over the observation window. Combining both terms, the total fractional variation in $\omega_{\rm app}$ over a window of duration Δt is

$$\left|\ln \frac{\omega_{\rm stop}}{\omega_{\rm guide}}\right| \le \left(4n_c + \frac{V_{\rm geo}}{d}\right)\Delta t \equiv \varepsilon, \tag{9}$$

where $n_c = \sqrt{\mu_\odot/r^3}$ is the local circular mean motion evaluated at the body's current heliocentric distance r. Table S5 evaluates ε numerically for representative orbital configurations with $\Delta t = 4\,\mathrm{h}$, the span between the midpoints of the guidance and stopping windows.

Table S5: Upper bound ε on $|\ln(\omega_{\rm stop}/\omega_{\rm guide})|$ for representative heliocentric orbits, $\Delta t = 4\,\mathrm{h}$.

Body type	$r_{\rm obs}$ (AU)	$d_{\rm obs}$ (AU)	$4n_c\,\Delta t$	$V_{\rm geo}\Delta t/d$
Outer planet (e.g. Jupiter at opposition)	5.2	4.2	0.0010	0.0011
Earth-crossing asteroid (near 1 AU)	1.0	1.0	0.0115	0.0069
Short-period comet (e=0.9, q=0.1 AU, at r=0.6 AU)	0.6	0.8	0.0247	0.0101
Sungrazer ($e\ge1$, q=0.1 AU, 20 d past perihelion)†	0.8	0.7	0.0160	0.0106

$V_{\rm geo}$ bounded by escape speed at r plus Earth's orbital speed; $n_c = \sqrt{\mu_\odot/r^3}$ evaluated at observed heliocentric distance. All $\varepsilon = 4n_c\Delta t + V_{\rm geo}\Delta t/d < 4\%$ for orbits satisfying the visibility criterion ($d \gtrsim 0.5\,\mathrm{AU}$, altitude $> 5°$).
†A sungrazer *at* perihelion ($r = q = 0.1\,\mathrm{AU}$) gives $4n_c\Delta t \approx 0.36$, but at this heliocentric distance the body is within $\sim 6°$ of the Sun and cannot satisfy the nighttime altitude criterion ($> 5°$).

Since $\varepsilon \ll 1$ for all orbits satisfying the visibility criterion (Table S5), Eq. (9) implies

$$\frac{\omega_{\rm stop}}{\omega_{\rm guide}} = 1 \pm \mathcal{O}(\varepsilon), \qquad \varepsilon \lesssim 4\%. \tag{10}$$

This is the slope-1 diagonal visible in Fig. 3: *any Keplerian body with geocentric distance $d \gtrsim 0.5\,AU$ appears to move at essentially constant apparent angular velocity over a four-hour observation window.*

The result follows directly from the conservation of specific angular momentum $h = r^2\,\mathrm{d}f/\mathrm{d}t =$ const (Kepler's second law). Although h is conserved about the *solar* focus while we observe from *Earth*, the correction is of order r_E/d and is already absorbed into the bounds above.

When can the diagonal be violated? The bound $\varepsilon \lesssim 4\%$ assumes $d \gtrsim 0.5\,\mathrm{AU}$. For a body to move *dramatically* slower during the stopping window than during guidance — the Matthew scenario — it must pass through an *apparent stationary point* (the transition from prograde to retrograde motion) within the four-hour window. At a stationary point the transverse geocentric velocity $V_\perp$ passes through zero, and $\omega_{\rm app}$ drops far below the guidance value.

Retrograde motion for outer planets occurs near opposition, when the planet rises in the east at sunset and transits near midnight. At the 5 BCE epoch, opposition directions lie well east of south — geometrically incompatible with the southward road corridor $[190°, 210°]$ required for the Magi's journey from Jerusalem to Bethlehem. For inner planets, retrograde occurs near inferior conjunction (near the Sun, hence not visible in the pre-dawn sky that satisfies the altitude criterion). In neither case can a stationary point coincide with the required azimuth and altitude window.

This is why the gold region (lower-right quadrant of Fig. 3) is empty not merely for the sampled orbits but for all Keplerian orbits: the diagonal is enforced by Kepler's second law, and the only mechanism that breaks the diagonal (a retrograde stationary point) is geometrically excluded by the azimuth and visibility constraints.

S1.9 Reference frames and the ground-observer constraint

The guidance verb προῆγεν (imperfect of προάγω) places motion in the observer's reference frame: the Star moved *before the Magi* as they walked. The stopping verb ἐστάθη (aorist passive of ἵστημι) similarly denotes apparent cessation of motion as perceived by an observer in Bethlehem. Both constraints must be evaluated in the local altitude-azimuth (AltAz) coordinate system of the ground observer, not relative to the stellar background.

The stopping-state transverse velocity requirement. The local AltAz frame co-rotates with Earth at the sidereal rate $\omega_{\mathrm{sid}} \approx 15^\circ\,\mathrm{h}^{-1}$. An object that appears *stationary* in this frame ($\omega_{\mathrm{local}} = 0$) must therefore move through the inertial ICRS frame at the local sidereal rate. At altitude h and azimuth az from a latitude ϕ observatory, the ICRS angular velocity of a "standing" object is

$$\omega_{\mathrm{stop}} = \omega_E \cos\phi \, \frac{\sqrt{\cos^2\mathrm{az} + \cos^2 h \, \sin^2\mathrm{az}}}{\cos(\mathrm{az} - \phi)}, \tag{11}$$

evaluating numerically to $14.76^\circ\,\mathrm{h}^{-1}$ at $h = 45^\circ$, $\mathrm{az} = 200^\circ$, $\phi = 31.7^\circ\mathrm{N}$ (Bethlehem), and ranging from 12°–$16^\circ\,\mathrm{h}^{-1}$ across the plausible altitude range (Table 1 of main text).

For any heliocentric body at geocentric distance d, the ICRS angular velocity is $\omega_{\mathrm{ICRS}} = V_\perp/d$ where $V_\perp$ is the transverse geocentric velocity. The stopping state therefore requires

$$V_\perp^{(\mathrm{stop})} = \omega_{\mathrm{stop}} \times d = 14.76^\circ\,\mathrm{h}^{-1} \times \frac{\pi}{180^\circ \times 3600\,\mathrm{s}} \times d. \tag{12}$$

Relationship to the stellar-background interpretation. A more permissive reading interprets the verbs as describing motion relative to the stellar background. In this frame the stopping criterion becomes $\omega_{\mathrm{proper}} \approx 0$, and the slope-1 diagonal analysis (§ S1.8) shows that guidance and stopping cannot both be achieved within the time span of the Jerusalem–Bethlehem journey. The corpus-linguistic results (Section 2 of SI) independently show that the stellar-background reading is not supported by the Greek: προῆγεν places the Star as an active agent leading the Magi (narrative register), not describing drift against the zodiac (astronomical register). The literal reading is therefore both more philologically defensible and, as shown above, more physically decisive.

S1.10 Geocentric orbital survey: detailed results

Unlike heliocentric objects (for which the primary impossibility is absolute), geocentric objects can in principle exhibit perceptible apparent motion followed by apparent deceleration, because both phases are governed by orbital velocity in the ground frame rather than by the sidereal rate. The slope-1 argument (§ S1.8) does not apply to geocentric orbits: near perigee of a highly elliptical orbit, ω_{app} can change by orders of magnitude over hours. A separate numerical survey is therefore required.

Table S6: Transverse geocentric velocity $V_\perp$ required for apparent stationarity in the ground frame ($\omega_{\text{stop}} = 14.76^\circ\,\text{h}^{-1}$) at representative geocentric distances. Since $V_\perp = \omega_{\text{stop}} \times d$, more distant objects require proportionally larger velocities.

Object / distance	d (AU)	$V_\perp^{(\text{stop})}$ (km s^{-1})	Fraction of c
Matney's comet during guidance window	0.004	42.9	0.014%
Near lunar distance	0.0026	27.8	0.009%
Near-Earth asteroid close approach	0.010	107	0.036%
Mars at opposition	0.52	5,560	1.85%
Jupiter at opposition	4.2	44,900	15.0%
Fastest solar system objects (sungrazers, $q \sim 0.005$ AU)	$V_{\text{orbital}} \lesssim 450\,\text{km}\,\text{s}^{-1}$		
Earth orbital speed	—	29.8	0.010%

Matney's comet: During the 08:00–10:00 guidance window ($d \approx 0.003$–0.004 AU, computed from Matney's published orbital elements: $q = 0.0407$ AU, $e = 1.0$, $T_{\text{peri}} =$ JD 1,719,785.6), apparent stationarity requires $V_\perp \approx 30$–$43\,\text{km}\,\text{s}^{-1}$—physically achievable, so the primary velocity impossibility does not apply to this case. Two separate failures rule it out instead. First, the comet is in full daytime sky throughout this window (local times 08:00–10:00, Sun above horizon), so naked-eye visibility is precluded regardless of orbital geometry. Second, the comet's proper motion against the stellar background is only $\omega_{\text{ICRS}} \approx 0.086^\circ\,\text{h}^{-1}$—negligible compared with the $\approx 15^\circ\,\text{h}^{-1}$ diurnal rate. Its apparent motion in the ground frame is therefore indistinguishable from any fixed background star: the "guidance" is the generic diurnal arc shared by every sky object, and the apparent azimuth "stopping" in Matney's Figure 6 is the meridian-transit geometric artifact that affects all objects near the meridian (§ S1.10), not a distinctive orbital behavior. **General rule:** For $d \gtrsim 0.1$ AU (required for naked-eye visibility when guidance motion is also apparent), $V_\perp \gtrsim 1{,}070\,\text{km}\,\text{s}^{-1}$—far above any known solar-system speed. This "primary impossibility" is independent of the transition dynamics analyzed in § S1.8.

The Moon. The Moon is the only natural geocentric body known in antiquity. Its specific angular momentum gives angular velocities of $0.61^\circ\,\text{h}^{-1}$ at perigee and $0.49^\circ\,\text{h}^{-1}$ at apogee — a factor of ~ 3 below the $2^\circ\,\text{h}^{-1}$ guidance threshold at every point in its orbit. The Moon never satisfies the guidance criterion.

Survey of hypothetical geocentric orbits. Script `sob_geocentric_survey.py` surveyed 28,800 geocentric orbit configurations with the following parameter ranges:

- Semi-major axis: 7,000 to 500,000 km (LEO to beyond lunar)
- Eccentricity: 0 to 0.99
- Inclination: all values (0°–180°)
- Argument of perigee ω, RAAN Ω, initial mean anomaly: full grid

Ground-frame apparent angular velocity (ω_{app}) was computed from topocentric altitude and azimuth using 5-minute timesteps. The Sun-altitude and object-altitude checks followed the same criteria as the heliocentric survey (Supplementary § S1.5), using a 12-hour nighttime window.

Results with the realistic stopping threshold ($\omega_{\text{stop}} < 0.20^\circ\,\text{h}^{-1}$, corresponding to less than one lunar diameter of apparent motion per hour): **zero configurations** satisfy guidance followed by stopping within 4 h in the azimuth corridor.

With the loose threshold $\omega_{\text{stop}} < 0.50^\circ\,\text{h}^{-1}$: two orbit families produce nominal hits, both equatorial ($i = 0^\circ$) at near-geosynchronous ($a \approx 42{,}164$ km, $e = 0.2$, period ≈ 24 h) and high-apogee ($a \approx 80{,}000$ km, $e = 0.5$, period ≈ 63 h) distances. In both cases the "stopping" value is 0.43–$0.46^\circ\,\text{h}^{-1}$ — approximately one lunar diameter per hour, detectable by naked-eye observation and not a credible "stood over" event. Moreover, the apparent slowing is a *geometric transit artifact*: as any equatorial object crosses the celestial meridian, the azimuth rate passes through zero (the object changes from moving west to moving east relative to south) while altitude continues to change, briefly minimizing the azimuth component of ω_{app}. This is identical to the transit effect analyzed for Matney's comet in the main Discussion. It is not an orbital deceleration.

Conclusion. No Earth-orbiting body — natural or hypothetical — can satisfy all criteria of the Matthew 2:9 narrative. The Moon fails the guidance criterion; all other hypothetical orbits either fail the stopping criterion or produce only the geometric transit artifact. Combined with the heliocentric result, the no orbit type satisfies the combined guidance, stopping, and azimuth criteria of Matthew 2:9.

S2 History of interpretation: detailed methods

S2.1 The unanimous pre-modern verdict: a supernatural, non-astronomical phenomenon

The foundational principle of historical philology is that a text should first be read as its earliest competent interpreters read it—those with native or near-native command of the source language, familiarity with its genre conventions, and relevant scientific knowledge. For Matt 2:9, this standard is met by the patristic commentators, who were fluent Greek readers trained in Hellenistic literary and scientific culture, and several of whom had direct acquaintance with the astronomy and astrology of their day. The survey of ancient and medieval readings in Allison (2005, pp. 18–34)[3] and Adair (2012)[4] reaches a conclusion that is both striking and historiographically important: *no commentator from the second century through the sixteenth century interpreted the Star as a natural astronomical event identifiable in the ordinary heavens.*

Augustine of Hippo (354–430 CE). Augustine is an especially significant witness because he came to Christianity after years as a Manichaean, during which he studied philosophy, astronomy, and the methods of the mathematici—professional astrologers and horoscope-casters (*Confessions* 4.1–3; 5.3). His familiarity with astronomical calculation was in fact part of what led him to reject Manichaeism: he found the astronomers' predictions accurate while the Manichaean cosmology was not (Ferrari 1973). Once a Christian, Augustine attacked stellar determinism vigorously. In his *Reply to Faustus the Manichaean* 2.6, he repudiated any astrological connection with the Star at Jesus's birth on explicitly astronomical grounds: the Star's behavior was manifestly incompatible with planetary motion, demonstrating it to be a miracle. The argument comes from someone who knew what planets actually do.

Origen of Alexandria (c. 185–253 CE). Origen produced the most extensive early Greek commentary on Matthew and the most sustained Christian intellectual engagement with Hellenistic astronomy and philosophy. His discussion of the Star in *Contra Celsum* 1.58–60 is sometimes misread as a naturalistic comet identification. In fact Origen stated only that the Star was *similar* to comets in its appearance and auspiciousness—not that it *was* a comet. The analogy is telling rather than naturalising: at the time, both Aristotle (*Meteorologica* 1.1) and the Alexandrian theologian Clement (known to Origen) classified comets as atmospheric, sublunar aberrations, not as celestial bodies in the heavens proper. To compare the Star to a comet was therefore to assimilate it to an anomalous, transient, earthly phenomenon, not to identify it with any regular feature of the sky. Origen's own theology of the event is unambiguous elsewhere: in *Homily on Numbers* 18.3–4, he stated that the Star *descended to Earth* just as the dove descended upon Jesus at his baptism—a trajectory that is the opposite of any orbital object and is structurally identical to his description of an angelic appearance. In *Commentary* *on Matthew* 13.6 and *Philocalia* 23.14 he argued against natal astrology; in *Contra Celsum* 1.60 he interpreted the Magi's original star-craft as demon-based, *destroyed* at the Incarnation. The Star was what replaced it, not what confirmed it. The view that Origen naturalised the Star is untenable.

John Chrysostom (c. 347–407 CE) and Gregory of Nyssa (c. 335–395 CE). Chrysostom, Archbishop of Constantinople and the most prolific early Greek expositor of Matthew, devoted the opening of his *Homilies on Matthew* to demonstrating that the Star was unlike any star or planet. Its motion over a short distance to a particular house—moving south from Jerusalem to Bethlehem, then stopping at street-level precision—was, he argued, evidence of a created angelic messenger, not a heavenly body. Gregory of Nyssa similarly emphasised the Star's anomalous behavior as proof of its supernatural character. Both theologians were writing in Greek as native speakers with access to Hellenistic astronomical and astrological literature.

Medieval authorities. In the Byzantine Empire, Emperor Manuel Comnenus (twelfth century) attempted to argue that astrological reasoning was the cognitive mechanism by which the Magi understood the Star's significance. The monk Michael Glycas rebutted him: crucially, even Manuel *agreed* that the Star itself was a new, miraculous object created by God; the dispute was solely about whether astrology was a licit intellectual framework for the Magi's interpretation of that miracle (Tester 1987, pp. 95–97)[5].

In the Latin West, Albertus Magnus, Roger Bacon, and Cardinal Pierre d'Ailly all engaged seriously with Arabic astrological tradition, constructing elaborate celestial frameworks for the Nativity. Yet all of them consistently distinguished the Star itself from the planetary configurations they were analyzing. D'Ailly stated on multiple occasions that the Star was miraculous, and he was explicit that astrological forces operate only within the domain of natural things; the supernatural Star stood outside that domain.

Kepler and his contemporaries. Johannes Kepler is frequently cited as the first astronomer to propose a naturalistic Star. In fact Kepler advanced a two-part hypothesis: the 7 BCE Jupiter–Saturn conjunction was a *sign* that prompted the Magi to travel, but the Star of Matt 2:9 was itself a distinct miraculous event. He wrote explicitly that it "was not of the ordinary run of comets or new stars, but by a special miracle moved in the lower layers of the atmosphere" (Burke-Gaffney 1937)[6]. Tycho Brahe expressed a similar judgment about celestial anomalies, and the Jesuit astronomer Giovanni Battista Riccioli (*Almagestum Novum*, 1651, 2:179–93) concurred: for all three astronomers, Matthew's Star was divine intervention, not an ordinary orbital body.

Protestant reformers. Martin Luther rejected astrological prognostication entirely and treated the Star as miraculous. John Calvin published a dedicated treatise against judicial astrology (1549), using the Star's manifestly non-planetary behavior as a reductio against the claim that Matthew described an astrological event.

The historiographical conclusion. The supernatural reading of Matt 2:9 is not a pious later overlay on a text that originally described something astronomical. It is the reading of every competent Greek-language interpreter for fifteen centuries, including individuals expert in the astronomical and astrological sciences of their day—the very sciences invoked in modern naturalistic proposals. The naturalistic reading is a post-Enlightenment innovation (c. 1800), motivated by philosophical commitments external to the text, not by a closer or more rigorous philological analysis of it.

S2.2 The modern naturalistic turn

Naturalistic readings of Matt 2 originate in the Enlightenment's program of rationalising biblical miracles: Hermann Samuel Reimarus (published posthumously 1774–1778) was one of the first to question the historicity of the Star, noting its miraculous character and its similarity to the pillar of fire in the Old Testament. The program of making the Gospels' extraordinary events into misunderstood natural occurrences was developed most systematically by Heinrich Paulus in the early nineteenth century; as applied to the Star, an early systematic expression was by the German Protestant theologian Christian Gottlieb Kühnöl (1807), who proposed a meteor shower, and by Friederich Münter (1820), who argued that the Star was the 7 BCE conjunction of Jupiter and Saturn. Münter thereby initiated the modern naturalistic search.

These readings were critically demolished by David Friedrich Strauss in his *Das Leben Jesu kritisch bearbeitet* (1835)[7]. Strauss demonstrated that the plain grammatical sense of Matt 2:9—the Star was *going before* the Magi (προῆγεν, imperfect: ongoing guided motion) *until* (ἕως) it stood over the house—could not be reconciled with any planetary body, and that the grammatical arguments used to avoid this conclusion (such as reading προῆγεν as a pluperfect) were linguistically inadmissible. Strauss also noted the structural parallel with Virgil's *Aeneid*, where a star guides Aeneas—the same legendary narrative register the present paper identifies statistically. By the early twentieth century, Albert Schweitzer (*Quest of the Historical Jesus*, 1913 ed.) dismissed the astrological interpretations of theologians such as Heinrich Voigt as "not worthy of a German professor of the twentieth century," and the philologist Franz Boll (1917)[8] pronounced the entire enterprise fruitless. After Strauss, academic biblical scholarship largely abandoned the naturalistic Star; by 1970 no peer-reviewed biblical journal was publishing defences of the Star as a physical object (Adair 2012).

The subject re-entered scientific journals with Hughes (1976)[9], publishing in *Nature*, who proposed the Jupiter–Saturn conjunctions of 7 BCE. Hughes's paper triggered renewed interest from astronomers and spawned a literature that now spans comets, novae, planetary conjunctions, and horoscopic events. The subsequent consensus among professional biblical scholars remained unchanged: the events of Matt 2 are theological narrative drawing on Numbers 24:17 and the conventions of ancient birth-legend, not a historical record amenable to astronomical identification (Brown 1993, pp. 610–613)[10].

S2.3 Attempts to re-read Matthew's Greek as astronomical or astrological terminology

Every naturalistic reading of Matt 2:9 requires some reinterpretation of the Greek vocabulary, since the plain reading—an object that moved as a guide before the Magi and then stopped over a specific house—is incompatible with the behavior of any known celestial body. Three successive attempts to supply a technical astronomical or astrological sense to Matthew's words can be distinguished.

Grammatical reinterpretation: Süskind and the pluperfect reading. Friedrich Gottlieb Süskind (early nineteenth century) argued that the imperfect verb προῆγεν should be read as a pluperfect—"had gone before"—removing any implication of active, ongoing guided motion during the journey. Strauss (1835, pp. 220–253)[7] refuted this definitively: the temporal particle ἕως ("until") governing the clause ἕως ἐλθὼν ἐστάθη ἐπάνω forces the preceding action (προῆγεν) to have been in progress up to the moment of stopping. A pluperfect reading is grammatically ruled out. This argument is reinforced by the imperfect aspect of προῆγεν itself, which in Koine Greek marks ongoing, durative action—precisely what a planet at a stationary point, by definition, is not exhibiting.

Lexical reinterpretation: Voigt and the heliacal rising. Heinrich Voigt (1911) proposed that the phrase ἐν τῇ ἀνατολῇ in Matt 2:2 and 2:9 should be translated not as "in the east" but as "at its heliacal rising"—a specific technical astronomical event. Franz Boll (1917)[8], then the leading authority on ancient Greek astronomical and astrological language, rejected this as naïve. The actual term used by ancient Greek astronomers for a heliacal rising was ἐπιτολή, not ἀνατολή. The TLG corpus search underlying the present paper (§ S3) confirms this: across 25 astronomical and astrological texts totalling approximately 622,000 words, ἀνατολή never functions as a technical designation for heliacal rising; it means simply "east" or generically "a rising." Albert Schweitzer (1913) dismissed Voigt's astrological case in its entirety, and Boll declared the enterprise of finding astronomical terminology in Matthew to be hopeless in the absence of the vocabulary that ancient astronomers actually used.

Systematic astrological reinterpretation: Molnar (1999). The most comprehensive recent attempt to read Matthew's vocabulary as technical astrological language is that of Michael Molnar[11], who proposed that the Star was a Jupiter–Moon occultation in Aries on 17 April 6 BCE, interpreted through Hellenistic horoscopic astrology. Molnar's linguistic claims require that several ordinary Greek words carry specialised astrological meanings.

(a) ἀνατολή as heliacal rising. Molnar independently follows Voigt in this reading. As established by Boll (1917)[8] and confirmed by the TLG survey in § S3, the technical astrological and astronomical term for heliacal rising was ἐπιτολή. Matthew uses ἀνατολή, which belongs to ordinary non-technical Greek. The present paper's corpus analysis finds zero instances of ἀνατολή functioning as a technical heliacal-rising term across the entire 25-text corpus.

(b) ἐπάνω ("over, above") as a zodiacal direction. Molnar interprets ἐστάθη ἐπάνω ("stood over") as indicating a favorable zodiacal configuration rather than a physical location above a house. The lexical evidence resists this: BDAG (s.v. ἐπάνω)[12] gives the primary sense as spatial superiority (above, over); in astronomical and cosmological texts it describes one sphere physically above another. It is not attested as a zodiacal directional indicator.

(c) ἐστάθη as a planetary stationary point. Molnar proposes that ἐστάθη (aorist passive of ἵστημι) encodes the technical concept of a planet reaching its synodic stationary point. This requires three independent failures to hold simultaneously: first, the relevant technical expression in Greek astronomical literature was not the verb ἵστημι alone but the noun phrase στηριγμός (station) with a copula, or the participle στηριζόμενος—forms the corpus survey in § S3 confirms Matthew does not use; second, the ἕως construction (§ S2.2) marks ἐστάθη as a punctual change of state from prior motion, which is not what a planetary station represents (a planet approaching its station is continuously decelerating over weeks, not moving and then stopping abruptly); third, a planetary station is a heliocentric phenomenon defined against the stellar background, whereas ἐστάθη ἐπάνω in context describes a position fixed above a terrestrial location. The Molnar reading thus requires simultaneously wrong lexical, grammatical, and cosmological mappings.

(d) Astrological geography: Aries as Judea. Molnar's hypothesis further requires that Aries was the astrological sign of Judea in the first century. This is contested in the ancient sources. While Ptolemy's *Tetrabiblos* 73–74 associates Aries with a broad Near Eastern region, Manilius (*Astronomica* 4.620–7, 797–8), writing in the early first century, assigns Palestine/Phoenicia to Aquarius; Dorotheus of Sidon assigns Gemini; other sources diverge further (Adair 2012, n. 6)[4]. Heilen (2015)[13], the most detailed recent study of Greco-Roman astrological geography, concludes that no stable first-century consensus assigned Aries specifically to Judea, and that the diversity of ancient assignments makes any single identification highly uncertain.

The philological verdict. J. Neville Birdsall (2002)[14]—a specialist in New Testament Greek and the textual transmission of the Greek New Testament, reviewing Molnar in the *Journal of the History of Astronomy*—identified multiple errors in Molnar's treatment of the Greek text that would not pass peer review in a philological context. These include misreadings of Greek morphology and the imposition of technical meanings onto vocabulary that ancient Greek sources consistently use in non-technical senses. Adair (2012)[4] catalogued further Greek errors across the Star-of-Bethlehem literature, including the proposal (in Teres 2002) that προάγω in Matt 2:9 is in the "aorist imperfect"—a tense that does not exist in Greek. The statistical corpus analysis in § S3 of this SI provides the systematic, quantitative complement to these individual philological observations: across 25 ancient Greek astronomical and astrological texts, Matthew's four key terms—προάγω, ἵστημι, ἐπάνω, and the verbal idea of celestial arrival for a specific person—are absent from astronomical usage at rates yielding a joint Bayes factor of ≈6,012 in favor of the legendary narrative register over the astronomical register. The ancient Greek writers who knew this vocabulary best never used it to describe what planets, comets, or stars do in the sky.

S3 Corpus linguistics: detailed methods

S3.1 Corpus construction and genre classification

The astronomical and astrological corpus was assembled from texts available in the Thesaurus Linguae Graecae (TLG) electronic database[15]. Texts were selected to represent all major genres and periods of ancient Greek astronomical writing from the fourth century BCE through the fourth century CE. Three genre classes were distinguished:

1. *Mathematical astronomy*: texts whose primary purpose is geometric or predictive description of celestial positions and motions—Ptolemy *Almagest* and *Planetary Hypotheses*, Hipparchus *Commentary on Aratus and Eudoxus*, Geminus *Isagoge*, Autolycus *On the Moving Sphere* and *On Risings and Settings*, Theodosius *Sphaerics* and *On Days and Nights*, Aristarchus *On the Sizes and Distances*, Cleomedes *Caelestia*, Hypsicles *Anaphorikos*.
2. *Astrological application*: texts that describe celestial phenomena in relation to human destinies—Ptolemy *Tetrabiblos*, Vettius Valens *Anthologies*, Hephaestion *Apotelesmatika*, Dorotheus *Carmen Astrologicum* (Greek fragments), Pseudo-Manetho *Apotelesmatica*, Paul of Alexandria *Eisagoge*. This genre is particularly important: astrological texts are the class most likely to have adopted Matthew's "relational" vocabulary of stars guiding or arriving for persons. Their failure to do so rules out a genre-based explanation of the statistical finding.
3. *Didactic and popularizing*: texts accessible to non-specialist ancient readers—Aratus *Phaenomena*, Eratosthenes *Catasterismi*, Achilles Tatius *Isagoge to Aratus*, Theon of Smyrna *Mathematics Useful for Reading Plato*, Aristotle *Meteorologica*, Theophrastus *De signis*.

S3.2 Corpus inventory (Table S7)

S3.3 Search and classification methodology

For each of the four lexical features (A–D; all words present in Matt 2:9), TLG morphological search was used to retrieve all inflected forms of the target lemma across the corpus. Feature E (ὁδηγέω/ἡγέομαι) was removed from the analysis because neither verb appears in Matthew's star-motion passage; testing words absent from the text under scrutiny is a category error. Occurrences were then classified by two independent criteria:

Table S7: Ancient Greek astronomical corpus used in the linguistic analysis.

Author	Work	Century	Word count	Genre
Ptolemy	*Almagest* (Syntaxis Mathematica)	II CE	185,000	Math. astr.
Ptolemy	*Tetrabiblos* (Apotelesmatika)	II CE	57,000	Astrol.
Ptolemy	*Planetary Hypotheses*	II CE	8,500	Math. astr.
Hipparchus	*Commentary on Aratus & Eudoxus*	II BCE	22,000	Math. astr.
Geminus	*Isagoge* (Intro. to Astronomy)	I BCE	18,000	Math. astr.
Autolycus	*On the Moving Sphere*	IV BCE	4,800	Math. astr.
Autolycus	*On Risings and Settings*	IV BCE	5,200	Math. astr.
Theodosius	*Sphaerics*	II BCE	9,000	Math. astr.
Theodosius	*On Days and Nights*	II BCE	4,500	Math. astr.
Aristarchus	*On the Sizes and Distances*	III BCE	5,500	Math. astr.
Cleomedes	*Caelestia* (Meteōra)	I CE	31,000	Math. astr.
Hypsicles	*Anaphorikos*	II BCE	3,500	Math. astr.
Aratus	*Phaenomena*	III BCE	10,000	Didactic
Eratosthenes	*Catasterismi*	III BCE	8,500	Didactic
Achilles Tatius	*Isagoge to Aratus*	III CE	7,500	Didactic
Theon of Smyrna	*Math. Useful for Reading Plato*	II CE	27,000	Didactic
Dorotheus	*Carmen Astrologicum* (Gk frags)	I CE	9,500	Astrol.
Pseudo-Manetho	*Apotelesmatica*	IV CE	14,000	Astrol.
Vettius Valens	*Anthologies*	II CE	82,000	Astrol.
Paul of Alex.	*Eisagoge*	IV CE	11,000	Astrol.
Hephaestion	*Apotelesmatika*	IV CE	43,000	Astrol.
Aristotle	*Meteorologica*	IV BCE	42,000	Didactic
Theophrastus	*De signis*	IV BCE	7,000	Didactic
Theon of Smyrna	*On Habitations* (Theodosius)	II BCE	2,800	Math. astr.
Hypsicles	*Anaphorikos* (2nd copy)	II BCE	3,500	Math. astr.
Total			**≈622,000**	

Word counts are approximate figures from TLG canon statistics and published critical editions. Exact counts depend on the edition and TLG version consulted and should be verified at the time of search. The qualitative conclusion—that Matthew's four vocabulary features are absent from or statistically marginal in the corpus—is independently documented in prior philological scholarship and is not contingent on precise word counts.

1. *Syntactic*: is the grammatical subject of the verb (or modified noun) a celestial body?
2. *Semantic*: does the usage match the specific sense found in Matthew 2:9 (spatial guidance of human travellers; spatial arrival of a star at a terrestrial point; locating a celestial body above a specific terrestrial address)?

Occurrences in grammatical or metalinguistic passages (e.g., Dionysius Thrax) were excluded. Classification was performed by the author against the explicit decision criteria documented in each feature definition above. All passages retrieved by TLG lemma search were individually read in context; every classification decision and its rationale are recorded at the passage level in Table S8 below, allowing independent verification by any reader with TLG access.

The following verbal and prepositional lemmas were searched:

- Feature A: προάγω — all inflected forms; classified for spatial/narrative guidance of a person toward a destination versus purely positional/temporal antecedence.
- Feature B: ἵστημι and στηρίζω — all inflected forms with celestial subjects; classified for

stationary-point technical usage versus other senses.

- Feature C: ἐπάνω and ὑπὲρ — all positional prepositional uses with celestial bodies; classified for locating a celestial body above a specific terrestrial point versus more general above-usage.
- Feature D: ἔρχομαι/ἐλθεῖν — all inflected forms with celestial subjects; classified for arrival at a terrestrial geographic point versus other spatial or temporal senses.

Feature E (ὁδηγέω and ἡγέομαι) was excluded from the statistical analysis because neither verb appears in Matthew's star-motion description (Matt 2:9). Ἡγέομαι occurs in Matt 2:6 referring to a messianic ruler, not the star; ὁδηγέω is absent entirely. Testing the frequency of words Matthew did not use would be a category error: the four retained features (A–D) test exclusively words Matthew *does* employ for the star's motion, giving a clean comparison with the corpus.

S3.4 TLG hit-level citation record

Table S8 records every passage retrieved by TLG lemma search for each of the four features. Authors returning zero hits for a feature are consolidated in a single row at the end of that section. The Matthew 2:9 sense against which each hit is classified is stated in the section header. "Neg" indicates the hit does not match the Matthew sense; "Pos" would indicate a match. No positive hit was found for any feature in any author.

Author (TLG ID)	Count	Locus	Usage in context	Pos?
Feature A: προάγω (lemma search). *Matthew sense*: star actively leads/guides human travellers toward a				
Ptolemy {0363}	3	*Tetrab.* 4.3.4.4; 3.14.25.6; 4.4.4.8	Noun cognates προαγωγαῖς, προαγωγικούς, προαγωγούς: astrological ranking/dignity terms ("promoting" a planet's influence), not directional guidance of a person.	Neg
Vettius Valens {1764}	4	*Anthol.* 1.20.123; 1.3.107; 8.7.67; *App.* 10.1.106	Astrological positional precedence (one planet "before" another in degree-order or timing); no human traveller, no destination.	Neg

Author (TLG ID)	Count	Locus	Usage in context	Pos?
Hephaestion {2043}	3	p.155 l.25; p.169 l.10 (*Apot.* 1); p.214 l.5 (*Apot.* 2)	Same positional/temporal astrological antecedence as Valens; no guidance motion toward a terrestrial goal.	Neg
Dorotheus {1337}	1	*Frag. Graeca* p.382 l.21	Astrological antecedence (planet "preceding" another in zodiac).	Neg
Cleomedes, Hipparchus, Geminus, Autolycus, Theodosius, Aristarchus, Hypsicles, Aratus, Eratosthenes, Achilles Tatius, Theon of Smyrna, Manetho, Paul of Alexandria, Aristotle (*Meteor.*), Theophrastus (*De signis*)	0 each	—	No instances of προάγω or cognates in any of these corpus texts.	Neg
Feature B: στηρίζω (lemma search; proxy for ἵστημι). *Matthew sense*: celestial body comes to rest above a				
Ptolemy {0363}	3	*Alm.* vol.1,2 p.463 l.11; *Hyp. plan.* vol.2 p.173 l.15; *Tetrab.* Bk 2 ch. 5 §2 l.5	Planet's first or second *stationary point* (στηρίζει) in its zodiacal cycle (direct→retrograde or retrograde→direct turning point); no terrestrial reference.	Neg
Hephaestion {2043}	10	*Apot.* 1: p.318 ll.5,9,12; *Apot.* 2: pp.58–59 (×4), p.109 l.5, p.304 l.15, p.337 ll.18,22	All 10 hits: technical στηρίζω for planetary stationary point in the zodiac; no celestial body stops "above" a house or person.	Neg
Dorotheus {1337}	4	*Frag. Graeca*: p.379 l.9; p.415 ll.15,19,21	Planetary stationary-point usage in natal or horary chart interpretation; zodiacal, not terrestrial.	Neg

Author (TLG ID)	Count	Locus	Usage in context	Pos?
Vettius Valens, Cleomedes, Hipparchus, Geminus, Autolycus, Theodosius, Aristarchus, Hypsicles, Aratus, Eratosthenes, Achilles Tatius, Theon of Smyrna, Manetho, Paul of Alexandria, Aristotle (*Meteor.*), Theophrastus (*De signis*)	0 each	—	No instances of στηρίζω in any of these corpus texts.	Neg
Feature C: ἐπάνω (lemma search). *Matthew sense*: celestial body localized ἐπάνω ("above/over") a specific				
Ptolemy {0363}	10	*Alm.* vol.1,1 p.475 l.3; vol.1,2 pp.40–44 (×4), pp.62–65 (×2), pp.152–153, pp.267–268	Geometric: "above" a construction line or reference arc in the *Almagest* mathematical diagrams; no celestial body, no terrestrial address.	Neg
Vettius Valens {1764}	6	*Anthol.* vol.4 p.149 l.28; 1.2.128; 1.18.90; 1.20.211; 5.1.49; *App.* 10.1.235	Astrological: planet or degree "above" a zodiacal sign or angular position; purely positional/relational, no terrestrial address.	Neg
Hephaestion {2043}	6	*Apot.* 1: p.96 l.10, p.314 l.14, p.333 l.15; *Apot.* 2: p.75 l.4, p.122 l.21, p.187 l.3	Astrological relational position (planet or lot "above" another in degree or dignity); no specific terrestrial point.	Neg
Theodosius {1719}	4	*De diebus et noctibus*: p.66 l.33; p.116 l.35; p.120 l.8; p.148 l.33	Geometric/temporal: arc or time interval "above" (i.e., exceeding) a reference value; no celestial body localized above a place.	Neg
Achilles Tatius {2133}	1	*Isag.* §18 l.28	Geometric: celestial sphere "above" the earth in a cosmographic description; not a specific body above a specific address.	Neg
Dorotheus {1337}	1	*Frag. Graeca* p.413 l.2	Astrological relational position.	Neg

Author (TLG ID)	Count	Locus	Usage in context	Pos?
Manetho {2583}	1	*Apoteles.* Bk 5 l.169	Astrological: planet above a sign or degree.	Neg
Aristotle {0086}	0 in corpus	(10 hits author-wide, none in *Meteor.*)	All Aristotle ἐπάνω hits are in works 001, 005, 006, 012, 014 (not *Meteorologica*); 0 in-corpus hits.	Neg
Theophrastus {0093}	0 in corpus	(10 hits author-wide, none in *De signis*)	All Theophrastus hits are in works 002, 004, 005, 010, 014, 017 (not *De signis*); 0 in-corpus hits.	Neg
Cleomedes, Hipparchus, Geminus, Autolycus, Aristarchus, Hypsicles, Aratus, Eratosthenes, Theon of Smyrna, Paul of Alexandria	0 each	—	No instances of ἐπάνω in any of these corpus texts.	Neg
Feature D: ἔρχομαι (lemma search). *Matthew sense*: a celestial body arrives at a specific terrestrial locatio				
Vettius Valens {1764}	6	*Anthol.* 5.7.86; 5.7.89; 8.5.65; 8.7.8; 8.7.26; 8.7.32	Planet "comes to" (ἐλθών) a zodiacal degree, sign, or aspect in natal or transit interpretation; the destination is a zodiacal locus, not a terrestrial address.	Neg
Hephaestion {2043}	1	*Apot.* 1: p.124 l.17	Same usage: celestial body's arrival at a zodiacal position.	Neg
Ptolemy, Cleomedes, Hipparchus, Geminus, Autolycus, Theodosius, Aristarchus, Hypsicles, Aratus, Eratosthenes, Achilles Tatius, Theon of Smyrna, Dorotheus, Manetho, Paul of Alexandria, Aristotle (*Meteor.*), Theophrastus (*De signis*)	0 each	—	No instances of ἔρχομαι with celestial subjects in any of these corpus texts.	Neg

S3.5 Thematic operationalization of features A–D

The four genre-register features A–D can be operationalized at two levels: *lexically* (does a specific Greek form appear with a celestial subject in the specified semantic role?) and *thematically* (does the underlying narrative event occur in the text, regardless of lexical choice?). These two operationalizations are correlated by definition—vocabulary and theme co-vary within genre—and are therefore not treated as independent Bayes factors. Their consistency is reported here as a robustness check: if the features measure genre-register signal rather than idiosyncratic word choice, both operationalizations should yield the same classification.

Table S9 shows the parallel formulations and the result of applying the thematic criterion to both corpora.

Table S9: Lexical and thematic operationalizations of features A–D, with results for the astronomical corpus and the guiding-star traditions.

Feat.	Lexical criterion	Thematic criterion	Astron. corpus	Guiding-star trad.
A	προάγω with celestial subject in narrative guidance of human travellers toward a destination	Celestial body actively directs human travellers toward a specific destination	0 (positional and temporal *precedence* only)	4 / 4 traditions
B	ἵστημι (not στηρίζω) with celestial body coming to rest at a specific location	Celestial body ceases motion at or above a specific terrestrial location (not a mathematical stationary point)	0 (στηριγμός is a zodiacal-longitude event in a geometric model; no body halts above a place)	2 / 4 traditions
C	ἐπάνω (not ὑπὲρ) locating a celestial body above a specific terrestrial address	Celestial body localized directly above a specific terrestrial address (building, person, plot of land)	0 (celestial bodies located above zodiacal regions or the abstract horizon, never above a terrestrial address)	0 / 4 traditions
D	ἔρχομαι/ἐλθεῖν with celestial subject arriving at a terrestrial geographic point	Celestial body arrives at a specific point on the Earth's surface	0 (arrival language for celestial bodies refers to zodiacal positions, aspects, or culminations; never to terrestrial points)	3 / 4 traditions

At every feature, the thematic operationalization yields the same result as the lexical operationalization for both corpora. The astronomical corpus scores zero at the thematic level for all four features, confirming that the absence of Matthew's vocabulary from astronomical texts reflects the absence of the corresponding *events*, not merely a difference in preferred vocabulary. The guiding-star traditions score identically at the thematic level as at the lexical level because those traditions were scored by narrative function from the outset. The Bayes factor derived from the lexical analysis (BF $\approx$ 6,012) therefore also characterizes the thematic evidence: the two analyses are measuring the same genre-register signal at different levels of abstraction.

Thematic assessment of the astronomical corpus. *Feature A (guidance of travellers).* The verb προάγω and its cognates appear in the astronomical corpus in two senses: temporal antecedence ("the epicycle *precedes* the deferent") and positional antecedence in longitude ("Mars *precedes* Jupiter by 15°"). In both senses the subject stands in a purely geometric or positional relationship to another celestial quantity; no text describes a star directing human agents toward a terrestrial destination. Astrological texts introduce human destinies as the *consequences* of planetary configurations but never as guided by a moving star. The narrative function—a celestial body leading travellers step by step toward a goal—is absent at every level of description.

Feature B (stopping above a location). Mathematical astronomy describes planetary stationary points (στηριγμός, the moment of zero longitudinal velocity relative to the fixed stars) as geometric events in the zodiac: a planet computed to have zero velocity in a given zodiacal longitude. The framework is wholly mathematical; no text in the corpus uses stationary-point language to say that a body "stands over" a house, city, or person. When the verb ἵστημι appears at all with a celestial subject (four instances in the corpus), it marks a different sense—the moment of computed position or of fixed configuration—not the cessation of motion above a terrestrial point.

The temporal particle ἕως ("until") in Matthew 2:9 further confirms that ἐστάθη denotes a *change of state*, not a prior static condition. The clause structure—"the Star was going before them [προῆγεν] *until* [ἕως] it stood [ἐστάθη] over the house"—makes ἐστάθη the temporal limit of the guidance action: the guiding motion ceased precisely when the standing began. In Koine Greek, ἕως governing an aorist indicative (ἕως ἐλθὼν ἐστάθη) marks the boundary at which the foregoing action terminated; a reading of the Star as *already* stationary throughout—making ἐστάθη merely descriptive of a pre-existing condition—is excluded by this boundary particle. The astronomical stationary-point reading (a planet at zero longitudinal velocity for a brief computed instant) is also grammatically incompatible with the imperfect προῆγεν (continuous ongoing guidance), which requires a sustained temporal extension that a mathematical instantaneous point cannot supply.

Feature C (localization above a terrestrial address). Spatial prepositions locate celestial bodies above zodiacal regions (ὑπὲρ Σκορπίου, "above Scorpius"), above the horizon (ὑπὲρ γῆν, "above earth"), or above other celestial bodies. The *referent* is always another celestial quantity or the abstract celestial sphere; no text localizes a star above a named building, named city, or named person in the sense of ἐπάνω οὗ ἦν τὸ παιδίον ("above where the child was"). This feature is absent thematically as thoroughly as it is lexically.

Feature D (arrival at a terrestrial point). Arrival language in the corpus marks celestial bodies reaching zodiacal positions, ingressing into signs, or completing synodic cycles. The referent of arrival is always a celestial coordinate; terrestrial arrival (a star *coming to* a specific house or person on the ground) is absent from all 25 texts.

Consistency with the lexical result. Because the lexical and thematic operationalizations agree at every feature for both corpora, the features A–D are functioning as genre-register markers rather than as idiosyncratic vocabulary tests. A critic who argued that Matthew's use of προάγω merely reflected an unusual but attested astronomical sense would still need to explain the thematic absence of guided-traveller narratives from the entire corpus; and a critic who argued that the guiding-star thematic comparison was too loose would still need to explain the lexical exclusion of Matthew's specific Greek forms from astronomical discourse. The two levels of evidence are mutually reinforcing without being statistically independent, and together provide a more robust basis for the genre-classification conclusion than either level alone.

S3.6 Statistical methodology

A one-sided Fisher exact test was applied to each 2×2 contingency table of the form:

	Matthew-type usage	Standard astron. usage
Matthew 2:9	a	b
Corpus	c	d

Because Matthew's star-motion passage comprises only ≈90 words, all Matthew-side cell counts are 0 or 1 ($b \in \{0, 1\}$, $a = 1 - b$). The individual Fisher exact probabilities are: $p_A = 0.100$, $p_B = 0.056$, $p_C = 0.033$, $p_D = 0.125$ (TLG-verified lemma counts, 2026-05-10; see Table S8). The Fisher combined chi-squared statistic is $X = -2 \sum_i \ln p_i = 21.35$ on df = 8, giving $p_{\text{Fisher}} \approx 6.3 \times 10^{-3}$. The product of individual probabilities—the correct joint data probability under independence, and the appropriate BF denominator—is $P(A, B, C, D \mid H_{\text{astro}}) = \prod_i p_i = 2.31 \times 10^{-5}$.

No feature individually passes Bonferroni threshold $\alpha/4 = 0.013$, but the joint Fisher combined statistic is significant ($p = 6.3 \times 10^{-3}$) and all four features contribute under the joint test.

Robustness to inter-feature correlation

The product formula $\prod p_i$ assumes independence of the four tests, which could be violated if certain text types tend to share multiple Matthew-like words simultaneously. To assess this, I modelled the joint null distribution using a Gaussian copula with equicorrelation parameter ρ: four standard normal variables $(Z_1, \ldots, Z_4)$ with correlation matrix $\Sigma_{ij} = \rho$ for $i \neq j$ were simulated, then converted to two-tailed p-values via $p_i = 2\Phi(-|Z_i|)$, and the Fisher combined statistic $X = -2 \sum_i \ln p_i$ was computed. The observed value $X = 21.35$ was compared against the simulated null distribution at each ρ using 2×10^6 Monte Carlo draws.

The corrected joint p remains below $\alpha = 0.05$ up to $\rho \approx 0.77$ (Table S10); at $\rho = 0.5$ the corrected $p = 0.028$. Brown's variance-corrected chi-squared method[16] gave consistent results throughout.

Table S10: Gaussian copula correction of the joint corpus p-value under equicorrelation ρ.

ρ	Corrected p (copula)	Brown's p
0.0	6.2×10^{-3}	6.3×10^{-3}
0.2	1.0×10^{-2}	8.7×10^{-3}
0.4	2.1×10^{-2}	1.7×10^{-2}
0.5	2.8×10^{-2}	2.4×10^{-2}
0.6	3.6×10^{-2}	3.2×10^{-2}
0.7	4.4×10^{-2}	4.1×10^{-2}
0.77	≈0.050	—
0.8	5.3×10^{-2}	5.1×10^{-2}
0.95	6.5×10^{-2}	6.5×10^{-2}

Equicorrelation ρ represents the average pairwise correlation among the four test statistics under H_0. A realistic upper bound for distinct lexical items across genre is $\rho \approx 0.5$; $\rho = 0.95$ would require near-identical distributional behavior for all four grammatically distinct feature classes. Script: `corpus_pvalue.py`.

Subset analysis (drop-one, $k = 3$ features): every drop-one-feature subset remains jointly significant at $\alpha = 0.05$ under independence (weakest drop-one: $p = 0.024$, dropping Feature C). The weakest three-feature subset yields $p = 0.067$ (dropping Features B and C together), which is not individually significant but reflects that B and C are the two most moderately-evidenced features; all other three-feature combinations give $p \leq 0.042$.

S3.7 Guiding-star corpus and Bayes factor (Table S11)

To contextualise the corpus-linguistic finding within the broader literary tradition, I assembled a comparative corpus of independently attested in contemporaneous or pre-Matthean guiding-star narrative *traditions*: each row represents one convention, not one text. Multiple texts reporting the same underlying event or sharing a known common source are collapsed to a single entry.

Collapsing rationale. Plutarch's *Timoleon* 8.7 and Diodorus Siculus 16.66.3 both describe the same Timoleon episode and almost certainly drew on the same earlier source (likely Timaeus of Tauromenium c. 345–260 BCE); treating them as independent data points would be double-counting one attestation. Plutarch describes the guiding phenomenon as a light from heaven (φέγγος οὐράνιον); Diodorus attributes it to the goddesses Demeter and Persephone, framing it as divine luminous guidance. LXX *Exodus* 13.21 and *Numbers* 14.14 represent a single continuous Israelite pillar-of-fire convention: *Numbers* is Moses recounting the Exodus tradition to God, not an independently invented guiding-star motif. *Wisdom of Solomon* 10.17 describes the same tradition as "a blazing flame of stars by night" (ἄστρων φλόγα), establishing its stellar characterization in Second Temple Jewish literature. In both cases, counting the texts separately would inflate the effective corpus size without adding independent evidential weight. The effect of collapsing (reducing N from 6 to 4) is to *raise* the Bayes factor slightly, because Laplace smoothing for the zero-count feature C is more generous at smaller N ($\mathrm{P(C)} = 1/6$ vs. $1/8$); the methodologically stricter corpus thus produces a modestly stronger result.

The *Protoevangelium of James* 21 is excluded because it is a derivative of Matthew 2:9–10, not an independent witness; its guiding-star language is adopted from Matthew. Plutarch's *Lysander* 12.1 and Cicero's *De Divinatione* 1.75 record the Dioscuri appearing alongside Lysander's fleet as an omen of victory at Aegospotami: they are stellar-epiphany texts with no directional guidance to a destination, and are excluded from the primary corpus but shown in the sensitivity analysis.

Table S11: Guiding-star narrative traditions: feature scores for Bayes factor analysis. Each row is one independently attested tradition. A = guidance verb; B = stopping verb (ἵστημι-type); C = ἐπι-based localization over a specific point (ἐπάνω or structural equivalent; ἐπί/ἐπάνω are interchangeable in the JosAsen manuscript tradition); D = arrival verb.

Tradition (constituent texts)	A	B	C	D
Primary corpus: four independent guiding-star traditions ($N = 4$)				
Virgil, *Aeneid* 2.687–711	1	1	0	1
Venus/Lucifer "marks the way" (signantemque vias); star terminates in Ida's forest (B); Aeneas arrives in Italy (D). No ἐπάνω localization. Servius (ad Aen. 2.694) identifies the celestial guide explicitly as Venus in her role as Lucifer (morning star), confirming this is a stellar *guiding tradition in Roman exegesis, not merely a light-omen.*				
Timoleon flame tradition	1	1	0	1
Plutarch, Tim. 8.7 + Diodorus 16.66.3 — same event, common source. Plutarch: "a light from heaven" (φέγγος οὐράνιον) runs with fleet (A), darts down on destination (B); Diodorus: light provided by the goddesses (A, D). Fleet arrives in Sicily (D).				
Lucian, *Navigium* 7; 9	1	0	1	1
Nav. 7 and 9 are explicit that the guide is a star (ἀστήρ); *Nav.* 7 calls it a saviour star and *Nav.* 9 describes its motions. Star first stands over (ἐπικαθίσαι) the masthead (τῷ καρχησίῳ) — the prefix ἐπι- is the same root as Matthew's ἐπάνω (ἐπί + ἄνω), and JosAsen manuscripts alternate ἐπί and ἐπάνω for the same angelic localization scene, confirming they are stylistic variants (C=1). Sequence: standing over → guiding (reverse of Matthew). Star then guides ship on its left side (ἐπὶ τὰ λαιά, A, D). No discrete stop at destination: B=0. Genre: satirical comic dialogue.				
Israelite pillar-of-fire tradition	1	0	1	0
LXX Exodus 13.21 + Numbers 14.14 — same convention. Pillar goes before (προπορεύεσθαι); ongoing guidance only, no discrete stop or arrival. ***C=1:*** *the cloud/pillar stands over (ἐπί) the Tabernacle in three explicit passages using the same preposition as Matthew's ἐπάνω: Exod 40.38 (νεφέλη γὰρ ἦν ἐπὶ τῆς σκηνῆς ἡμέρας καὶ πῦρ ἦν ἐπ' αὐτῆς νυκτός), Num 9.15 (καὶ ἑσπέρας ἦν ἐπὶ τῆς σκηνῆς ὡς εἶδος πυρὸς ἕως πρωί), and Num 9.18–22 (πάσας τὰς ἡμέρας, ἐν αἷς σκιάζει ἡ νεφέλη ἐπὶ τῆς σκηνῆς). As established in the table header, ἐπί and ἐπάνω (ἐπί + ἄνω) are structural equivalents for localization above a specific point; the pillar tradition therefore satisfies feature C. Wisdom of Solomon 10.17 (same tradition) describes the pillar as "a blazing flame of stars" (ἄστρων φλόγα), establishing the stellar characterization of this tradition in Second Temple Jewish literature and confirming it belongs to the divinely-sent luminous guide category rather than a purely fire-omen category.*				
Stellar-omen texts (sensitivity analysis only; excluded from primary corpus)				
Josephus, *Jewish War* 6.289	0	1	0	0
Sword-like star ἔστη over Jerusalem (B=1, same verb as Matthew) but uses ὑπὲρ not ἐπάνω (C=0); labeled τερατεία; no guidance (A=0) or arrival (D=0).				
Dio Cassius, *Roman History* 54.29.8	0	0	0	0
Comet over Rome at death of Agrippa; verb αἰωρηθείς (NOT ἵστημι); prep. ὑπὲρ (NOT ἐπάνω). No guidance or arrival.				
Plutarch, *Lysander* 12.1	0	0	0	0
Dioscuri appear alongside fleet as victory omen; no guidance, no destination.				
Cicero, *De Divinatione* 1.75	0	0	0	0
Same Aegospotami/Dioscuri tradition.				
Matthean derivative (excluded)				
Protoevangelium of James 21	—	—	—	—
Paraphrases Matt 2:9–10 verbatim; not an independent witness.				
Supernatural descent texts (feature C commentary only; not in BF corpora)				
PGM I.54–95	0	1	0	1

Continued on next page

Table S11 continued

Tradition (constituent texts)	A	B	C	D
Descending star-god stands ἐν μέσῳ τοῦ δώματος ("in the middle of the housetop") — ἐν μέσῳ, not ἐπάνω (C=0). No guidance (A=0).				
Joseph and Aseneth 14.4 (Burchard longer rec.)	0	1	1	1
Morning star descends as divine messenger (ἄγγελος); stands ἔστη ἐπάνω τῆς κεφαλῆς αὐτῆς (C=1). Departs eastward in solar chariot (star = messenger identity). Philonenko shorter rec.: ἐπί (C=0). Genre: diaspora Jewish novel *— same broad literary type as the gospels. No guidance (A=0).*				
Feature frequency (primary, $N = 4$)	**4/4**	**2/4**	**2/4**	**3/4**
$P(\text{feat.} \mid H_{\text{guide}})$, Laplace $k = 1$	5/6	3/6	3/6	4/6

Joint: $P(A, B, C, D \mid H_{\text{guide}}) = \frac{5}{6} \times \frac{3}{6} \times \frac{3}{6} \times \frac{4}{6} \approx 1.39 \times 10^{-1}$. BF $= 1.39 \times 10^{-1}/2.31 \times 10^{-5} \approx 6{,}012$ (exceeds 100 on the Jeffreys scale). $P(A, B, C, D \mid H_{\text{astro}}) = \prod_i p_i = 2.31 \times 10^{-5}$ (product of individual Fisher exact probabilities; the Fisher combined tail-p of 6.3×10^{-3} is a meta-test statistic, not a joint data probability). Sensitivity: +Josephus & Dio ($N = 6$): BF = 2,537; +all omen ($N = 8$): BF = 1,040; $k \in [0.5, 3]$: BF $\in$ 4,546–6,818; 3× denominator uncertainty: 2,004–18,038. **Feature C key finding:** ἐπι-based localization (ἐπάνω or structural equivalent) scores 0 in all astronomical texts, all omen texts, and PGM I.54–95. It appears in exactly three non-Matthean texts: Lucian, *Nav.* 9 (ἐπικαθίσαι τῷ καρχησίῳ; a comic dialogue); *JosAsen* 14.4 longer rec. (ἐπάνω τῆς κεφαλῆς; a diaspora Jewish novel); and the Israelite pillar-of-fire tradition, where ἐπί marks the cloud/pillar standing over the Tabernacle at Exod 40.38, Num 9.15, and Num 9.18–22. JosAsen manuscripts alternate ἐπί and ἐπάνω for the same scene, confirming these are stylistic variants in the narrative register. All three attestations belong to the legendary or sacred-narrative register; the astronomical and omen corpora use ὑπὲρ exclusively. Script: `guiding_star_bayes.py`.

S4 Software and computational environment

All orbital mechanics computations and statistical analyses were performed in Python 3.12. The principal libraries used were:

- `astropy` v6.x[17] — coordinate transformations, JPL ephemeris access, time systems.
- `NumPy`[18] — vectorised array operations and `einsum`-based orientation sweeps.
- `SciPy`[19] — Fisher's exact test (`scipy.stats.fisher_exact`), Newton–Raphson utilities.
- `matplotlib`[20] — all figures.

Code and data necessary to reproduce all figures and tables in the main paper and this Supplementary Information are available at `https://github.com/adairaar/sob_astro_test`. The repository includes:

- `sob_proof_fine.py` — kinematic demonstration and exhaustive heliocentric orbital survey (5° angular grid, 529 million configurations)
- `sob_mc_refined.py` — Monte Carlo refinement sweeps in continuous parameter space around the best-fit grid orbit and Matney's proposed 5 BCE comet; 2.25 million configurations sampled, zero satisfying all criteria
- `sob_epoch_robustness.py` — multi-epoch survey across all proposed Star of Bethlehem dates (12 BCE, 7 BCE, 5 BCE, 1 BCE), recomputing Earth positions and ENU matrices from the JPL ephemeris at each date; total 79,788 visible configurations across 7 epochs
- `sob_calcs.py` — Matney comet orbital analysis
- `sob_figures.py` — figure generation
- `corpus_pvalue.py` — Gaussian copula simulation and Brown's method for joint p-value robustness; 2×10^6 Monte Carlo draws per correlation level

- `corpus_linguistics.py` — Fisher's exact test and contingency tables
- `data/` — parameter grids, survey outputs, corpus counts

S5 Supplementary discussion

S5.1 Corpus-linguistic finding in context

The linguistic evidence adds a dimension the mechanics alone cannot supply. Even granting an unknown physical mechanism capable of producing the required velocity change, such a mechanism would still leave unexplained why an evangelist reporting on a celestial phenomenon used none of the vocabulary that Greek astronomical and astrological writers used for exactly that purpose over eight centuries of writing. The vocabulary of leading (προάγω), arriving (ἔρχομαι), and standing over (ἵστημι ἐπάνω) belongs to the narrative register that Greek authors used for persons and supernatural agents. This has been observed qualitatively in prior scholarship[4, 21]; the present analysis quantifies it with a Fisher combined $p \approx 6.3 \times 10^{-3}$ against the null hypothesis that Matthew's usage is drawn from the same distribution as the astronomical corpus. When the alternative hypothesis is framed as the well-attested guiding-star narrative tradition of Greco-Roman and Jewish literature, the Bayes factor in favor of the narrative reading is BF $\approx$ 6,012 (exceeding 100 on the Jeffreys scale), dropping to 1,040 under the most conservative sensitivity analysis that pools guiding and omen texts together.

A potential methodological concern is that the lexical analysis (Fisher's exact tests against the astronomical corpus) and the Bayes factor comparison operate at different levels of abstraction: the former compares specific Greek vocabulary against a Greek-language corpus, while the latter scores narrative traditions thematically, across multiple languages. These are not inconsistent but complementary, because features A–D are genre-register markers at both levels simultaneously. The thematic robustness check (Table S9) confirms that the null result holds at the thematic level: Greek astronomical and astrological texts do not describe celestial bodies guiding human travellers toward a destination, ceasing motion above a specific house, or arriving at a terrestrial address—at any level of description. The lexical and thematic analyses converge on the same genre classification. Because lexical and thematic features co-vary within genre by definition, the two are treated as convergent confirmation rather than independent Bayes factors; their consistency confirms the genre-register signal.

The register of Matthew 2:9, on this dual-level analysis, is that of ancient narrative prose—the language of novels, legendary accounts, and divinely-guided journeys—rather than of astronomical or astrological writing. This conclusion holds whether the evidence is drawn from specific Greek vocabulary or from the narrative functions the text describes.

S5.2 ἐπάνω and the preposition data in detail

Feature C (ἐπάνω) is particularly diagnostic of literary register. In *Jewish War* 6.289, Josephus describes a sword-shaped star that "stood" (ἔστη, the same ἵστημι verb as Matthew, feature B) "over" the city, but uses the preposition ὑπὲρ rather than ἐπάνω; he explicitly classifies the portent as a τερατεία ("prodigy") rather than an astronomical observation. Dio Cassius uses αἰωρηθείς ("suspended/lifted") and likewise ὑπὲρ. Every astronomical and omen text without exception uses ὑπὲρ for a celestial body positioned above a terrestrial point; not one uses ἐπάνω or any ἐπι-compound in that role.

By contrast, ἐπι-based localization is attested in the guiding-star narrative tradition: Lucian's *Navigium* 9 uses ἐπικαθίσαι τῷ καρχησίῳ ("stood upon the masthead") and the *Joseph and Aseneth* 14.4 longer recension (Burchard) uses ἔστη ἐπάνω τῆς κεφαλῆς αὐτῆς for an angelic figure; crucially, the Philonenko shorter recension of the same passage substitutes ἐπί, showing that ἐπί and ἐπάνω were treated as interchangeable by scribes working within the narrative register. The genre of *Joseph and Aseneth* is significant: it is a diaspora Jewish novel, legendary narrative prose of the same broad literary kind as the Gospels, not an astronomical treatise or omen record. Feature C therefore discriminates not between two equal options but between two distinct literary registers: ὑπὲρ in observational and omen discourse; ἐπάνω/ἐπι-compounds in legendary narrative texts.

The closest parallels for a star-like divine entity descending to ground level are the Greek Magical Papyri and *Joseph and Aseneth*. In PGM I.54–95 a spell summons a god in the form of a star to come down (κατελθών) to the earth (ἔγγαιος); the star stands ἐν μέσῳ τοῦ δώματος ("in the middle of the housetop"), not ἐπάνω. In *Joseph and Aseneth* 14.1–5 the morning star descends, manifests as a radiant heavenly visitor, and departs eastward before sunrise; the longer recension uses ἐπάνω, the shorter uses ἐπί[21]. Neither text involves guidance of travellers (features A, D), and neither enters the Bayes factor corpora. They do confirm that ἐπι-based localization for a star-entity belongs exclusively to the legendary narrative register, not to observational or omen writing.

S5.3 Legendary genre of the guiding-star tradition

Every tradition in the primary guiding-star corpus belongs to a legendary, mythological, or openly non-historical genre. Virgil's *Aeneid* is mythological epic; the Timoleon flame is legendary history that ancient authors themselves treated as miraculous embellishment; Lucian's *Navigium* is a satirical comic dialogue; and *Joseph and Aseneth* is a diaspora Jewish novel. Matthew's linguistic profile aligns most closely with these legendary and mythological narratives—not with the sober observational language of ancient astronomy. This is positive confirmation of the Bayes factor result: the narrative register in which Matthew 2:9 is most at home is the legendary narrative register, not that of astronomical observation.

S5.4 Within-Matthew lexical consistency analysis

The corpus Bayes factor ($BF_{corpus} \approx 6{,}012$) tests whether Matthew 2:9 belongs to the astronomical or the guiding-star narrative register by comparing those two external corpora drawn from the TLG and the guiding-star tradition. An independent second Bayes factor can be derived from an entirely different dataset: Matthew's own Gospel. If Matthew himself never uses any of the four words in the senses required by the astronomical reading, then Matthew's own idiolect provides an independent prior that any astronomical reading of Matt 2:9 is highly improbable. Because the two datasets (Matthew's Gospel and the TLG corpus) are completely disjoint, the two Bayes factors are treated as conditionally independent and their product is a valid joint Bayesian update.

Independence of the two analyses. BF_{corpus} is computed from the TLG astronomical corpus (≈622,000 words in 25 texts) and the four guiding-star traditions; Matthew's Gospel contributes no data to that calculation. $BF_{Matthew}$ (derived below) is computed entirely from Matthew's 27 other chapters; the TLG corpus contributes no data. The two datasets share no texts and test related but distinct aspects of the same hypothesis: BF_{corpus} asks whether Matt 2:9's vocabulary pattern is characteristic of astronomical or legendary narrative discourse in general, while $BF_{Matthew}$ asks whether Matthew's own vocabulary habits elsewhere make an astronomical reading of Matt 2:9 plausible. Both answer the same overarching question (does Matt 2:9 belong to astronomical or legendary narrative discourse?) from independent evidence, so their product gives the combined posterior odds.

Note that lexical evidence (specific word forms) and narrative evidence (the events described) are separable in principle: a writer can use non-technical vocabulary to describe astronomical events, and can use astronomical vocabulary metaphorically for terrestrial events (e.g., "retrograde motion" applied to cars being overtaken on a motorway). That Matthew's text fails the astronomical hypothesis at *both* levels—non-astronomical vocabulary (features A–D) *and* non-astronomical events (thematic operationalization, Table S9)—from two independent datasets constitutes a double-hit that cannot be attributed to a single shared genre signal.

Data and BF computation. To test this, I compiled every occurrence of each of the four features (προάγω, ἵστημι, ἐπάνω, ἔρχομαι) in Matthew's Gospel outside Matt 2:9, verified against the BibleHub interlinear (Strong's numbers G4254, G2476, G1883, G2064 respectively; searches cross-checked against NA28/SBLGNT lemmatization) and applied the same classification criteria used in the TLG corpus analysis.

For each feature i, let n_i be the number of other Matthean occurrences and $k_i = 0$ the number in which an astronomical sense is attested. Using Laplace smoothing, the probability of an astronomical sense in Matt 2:9 under H_{astro} is estimated from Matthew's own idiolect as $p_i^{\text{astro}} = (k_i + 1)/(n_i + 2) = 1/(n_i + 2)$. Under H_{guide}, the probability of the observed narrative sense is $p_i^{\text{guide}} = (n_i + 1)/(n_i + 2)$. The per-feature Bayes factor from within-Matthew data is therefore

$$\mathrm{BF}_i^{\text{Matthew}} = \frac{p_i^{\text{guide}}}{p_i^{\text{astro}}} = \frac{(n_i + 1)/(n_i + 2)}{1/(n_i + 2)} = n_i + 1. \tag{13}$$

The within-Matthew Bayes factor for each feature is thus simply the count of non-astronomical uses in Matthew's Gospel plus one.

Syntactic note on προάγω (Feature A). Προάγω is ambitransitive: it can be used transitively (with a direct object, meaning "to lead / guide someone") or intransitively (meaning "to go before / precede"). In Matt 2:9 the verb is *transitive*: προῆγεν αὐτούς, "was guiding them," with the Magi as explicit direct object. This is the active-guidance, agent-leading construction; it does not admit the "star merely preceding them in the sky" reading, because that reading requires the intransitive form. All five comparanda outside Matt 2:9 are also transitive with a human direct object and describe active spatial direction toward a named destination (Matt 14:22: disciples to the other side; Matt 21:9×2: crowds before Jesus; Matt 26:32, 28:7: Jesus going ahead of the disciples to Galilee). The intransitive NT uses of προάγω tend toward figurative or temporal antecedence; the transitive-with-human-direct-object construction is consistently and exclusively spatial navigation in Matthew's usage. Notably, Matthew never employs προηγέομαι, the intransitive compound that most naturally expresses positional antecedence ("going before" without a direct object), which is what the astronomical reading would require. Its absence reinforces that Matthew's motion vocabulary in Matt 2:9 is not positional-antecedence language.

Available alternative vocabulary for ἵστημι (Feature B). In Greek astronomical and astrological texts, the standard term for a planet reaching its stationary point (the moment of apparent reversal in the zodiac) is στηρίζω and its cognates, not ἵστημι. Στηρίζω appears 17 times in the TLG corpus with this technical meaning; Matthew never uses it anywhere in his Gospel. If Matthew 2:9 intended an astronomical stationary point, the expected vocabulary was available and familiar to Hellenistic writers: στηρίζω, not the generic ἵστημι of persons stopping at physical locations. Matthew's consistent avoidance of the astronomical term, and his exclusive use of ἵστημι for persons and objects at concrete terrestrial places, is precisely the pattern expected under the legendary narrative reading and unexpected under the astronomical reading.

Physical-contact force of ἐπάνω (Feature C). Matt 28:2 is the key Matthean comparandum: the angel of the Lord descends and sits ἐπάνω the stone—a heavenly being in point-contact with a physical object. This is the usage register of Matt 2:9 (ἐπάνω οὗ ἦν τὸ παιδίον, "above where the child was"), where a divine entity is localized to the building-level. Neither the angel's position in Matt 28:2 nor the star's position in Matt 2:9 describes an astronomical altitude or zodiacal latitude; both locate a heavenly entity at a specific, named terrestrial point. This is consistent with, and strongly supports, the TLG finding that ἐπάνω with celestial/supernatural subjects belongs to the legendary narrative register.

The expected preposition for an astronomical body positioned above a terrestrial region is ὑπέρ ("over", in the sense of spatial elevation without contact), which is used consistently in the TLG corpus when a celestial body is described above a city, region, or zodiacal boundary. Every astronomical and omen text in the corpus that places a celestial body above a terrestrial point employs ὑπέρ, never ἐπάνω. Read literally, ἐπάνω alone therefore semantically disqualifies the astronomical reading: a star cannot be "over" a specific house in the sense of point-contact or immediate surface proximity. The $\mathrm{BF}_{\mathrm{Matthew}}$ calculation is nevertheless deliberately conservative: it counts only occurrence frequencies within Matthew's idiolect, not the semantic force of the words themselves. The semantic argument from ἐπάνω versus ὑπέρ is an independent, qualitative strengthener that is *not* folded into the quantitative Bayes factor.

Evidence force of Feature D. Ἔρχομαι is the most frequent verb in Matthew's Gospel ($\approx$87 occurrences), which means one could argue that even an astronomical description would happen to use it by chance. The within-Matthew evidence refutes this reasoning: Matthew does not use ἔρχομαι of celestial bodies reaching celestial positions in any of his 86 other uses. Its overwhelmingly consistent referent is the arrival of a person or agent at a terrestrial location—exactly the sense in Matt 2:9. The verb's frequency strengthens rather than weakens the within-Matthew inference: with 86 non-astronomical comparanda and zero uses in an astronomical positional sense, the probability that Matt 2:9 is drawing on an astronomical sense is $\leq 1/88$ by Laplace smoothing. Feature D's high n makes its p the smallest of the four, and excluding it from the analysis would be methodologically unjustified: one does not discard evidence because it is particularly strong for one side. Additionally, Matthew never uses καταντάω ("to arrive at", "to reach"), which appears in Hellenistic astronomical contexts for a body reaching a celestial coordinate or completing a synodic phase. Its availability but non-use by Matthew further supports the conclusion that his arrival vocabulary is consistently confined to the movement of persons to terrestrial locations.

Combined $\mathbf{BF}_{\mathbf{Matthew}}$. Assuming independence of the four features within Matthew's idiolect (reasonable for distinct lexical items across different grammatical categories), the joint within-Matthew Bayes factor is the product of the per-feature factors:

$$\mathrm{BF}_{\mathrm{Matthew}} = \prod_{i\in\{A,B,C,D\}} \mathrm{BF}_i^{\mathrm{Matthew}} = 6 \times 21 \times 8 \times 87 = 87{,}696. \tag{14}$$

This factor quantifies how much more probable Matthew's own vocabulary habits (across his full Gospel) are under H_{guide} than under H_{astro}: an author writing legendary narrative prose would naturally use all four words in the senses consistently attested throughout Matthew, whereas an author writing astronomical description would be expected to use them in astronomical senses that Matthew never employs elsewhere.

Joint Bayes factor combining both datasets. Because $\mathrm{BF}_{\mathrm{Matthew}}$ and $\mathrm{BF}_{\mathrm{corpus}}$ are derived from entirely disjoint datasets (Matthew's Gospel versus the TLG corpus and guiding-star traditions), they are treated as conditionally independent, as they derive from non-overlapping corpora and distinct sampling frames. Their product is therefore the joint Bayes factor incorporating the linguistic evidence considered here:

$$\mathrm{BF}_{\mathrm{joint}} = \mathrm{BF}_{\mathrm{Matthew}} \times \mathrm{BF}_{\mathrm{corpus}} = 87{,}696 \times 6{,}012 \approx 5.3 \times 10^8. \tag{15}$$

The conservative lower bound on $\mathrm{BF}_{\mathrm{corpus}}$ (1,040, from the sensitivity analysis pooling guiding and omen texts) gives a conservative joint lower bound:

$$\mathrm{BF}_{\mathrm{joint}}^{\mathrm{conservative}} = 87{,}696 \times 1{,}040 \approx 9.1 \times 10^7. \tag{16}$$

The two Bayes factors address related but distinct questions. $\mathrm{BF}_{\mathrm{corpus}}$ asks: given two external reference corpora (TLG astronomical texts and the guiding-star traditions), to which is Matt 2:9's vocabulary pattern more similar? $\mathrm{BF}_{\mathrm{Matthew}}$ asks: given Matthew's own lexical habits across his Gospel, how much more probable is the narrative reading than the astronomical reading of Matt 2:9? The fact that two independent datasets—one external to Matthew's Gospel, one internal to it—converge on the same conclusion strengthens the genre classification beyond what either could show alone. Notably, the combined result is robust to the feature-independence assumption within each BF: even if inter-feature correlation within the TLG analysis is as high as $\rho = 0.77$ (where $\mathrm{BF}_{\mathrm{corpus}}$ remains significant; see copula sensitivity table), the joint factor $87{,}696 \times \mathrm{BF}_{\mathrm{corpus}}(\rho{=}0.77)$ still enormously exceeds any conventional threshold.

For completeness, the frequentist summary of the within-Matthew evidence is the Fisher combined statistic: $X = -2\sum_i \ln(1/(n_i + 2)) = 23.42$, df $= 8$, $p \approx 3 \times 10^{-3}$. This confirms the direction of the Bayesian result but, as with any p-value, understates the magnitude of the evidence because it does not incorporate the numerator of the likelihood ratio.

S5.5 Nighttime close-flyby comet survey and optimizer

Motivation. The primary-impossibility argument of § S1.9 shows that for $d \gtrsim 0.1\,\mathrm{AU}$ the required transverse velocity $V_\perp$ for the stopping phase exceeds any known solar-system speed. For the very close-flyby regime ($d \sim 0.003\,\mathrm{AU}$), $V_\perp \sim 30\,\mathrm{km\,s^{-1}}$ is physically achievable and the sidereal-frame argument closes differently (Matney's comet is in daylight; apparent motion in the ground frame is indistinguishable from any fixed star). However, a loophole remains: a nighttime comet at $d \sim 0.001$–$0.01\,\mathrm{AU}$ on a different orbit might, in principle, exhibit the required guidance-then-stopping behavior even though Matney's specific solution fails. This section closes that loophole with a direct Monte Carlo survey and a complementary numerical optimizer.

Survey design. Script `sob_close_flyby_mc.py` (available in the online repository) sampled $N_{\mathrm{MC}} = 10{,}000{,}000$ heliocentric orbits. Orbital parameters:

- Perihelion distance: $q \in [0.30, 1.40]\,\mathrm{AU}$ (uniform).
- Eccentricity: stratified across three groups—high-elliptic ($e \in [0.70, 1.00)$, one-third of orbits), near-parabolic ($e \in [1.00, 1.01)$, one-sixth), and hyperbolic ($e \in [1.01, 5.00)$, one-half)—to ensure dense sampling near $e = 1$ where cometary orbits concentrate.
- Inclination: stratified to concentrate sampling in the low-inclination zone that first principles and the prior optimizer result ($i = 2.2°$) indicate as the only viable regime. Specifically: 60% of orbits draw i uniformly in $[0°, 20°]$ (prime candidate zone); 25% in $[20°, 45°]$; 15% in $[45°, 180°]$ (full-sky tail, retained for the probabilistic statement and completeness). Each stratum uses $\cos i$ uniform within its range for correct angular coverage.
- Longitude of ascending node Ω and argument of perihelion ω: uniform on $[0°, 360°)$.
- Perihelion epoch offset ΔT: uniform on $[-90, +90]$ days from JD 1 719 755.898 (the primary event date, 5 BCE Jun 8).

Evaluation proceeded in two phases. Phase 1 (quick reject): a vectorized parabolic approximation computed the minimum geocentric distance $d_{\min}$; orbits with $d_{\min} > D_{\mathrm{flyby}} = 0.02\,\mathrm{AU}$ were rejected (of 2 million orbits, 54 survived). Phase 2 (detailed evaluation): exact orbital mechanics (Barker's equation for $e = 1$; Newton–Raphson on the Kepler / hyperbolic mean-anomaly equation otherwise) with analytic AltAz transformation computed the apparent angular velocity at 5-minute intervals across a 7.7-hour nighttime window (Sun below $-6°$). Criteria:

1. *Night visibility:* Sun altitude $< -6°$ and comet altitude $> 10°$.
2. *Road-corridor guidance:* azimuth within $[190°, 210°]$, azimuth drift $|daz/dt| < 1.5° \,\mathrm{h}^{-1}$, for a contiguous window of ≥ 1.0 h.
3. *Stopping:* total apparent angular velocity $\omega_{\mathrm{app}} < 0.30° \,\mathrm{h}^{-1}$ within 4 h of the guidance window.

Survey results. Of 10 million orbits, 1,041 survived the quick-reject phase (geocentric distance $< 0.05\,\mathrm{AU}$ after parabolic approximation); 66 were genuine close flybys after exact integration ($d_{\min} < 0.02\,\mathrm{AU}$). **Zero orbits satisfied all three criteria simultaneously.** All 66 genuine-flyby orbits failed criterion (ii): no contiguous guidance window of $\geq 1\,\mathrm{h}$ was achieved at the correct azimuth. The stratified inclination sampling concentrated $\approx 85\%$ of draws in $i < 45°$, ensuring thorough coverage of the geometrically viable region while retaining the full-sky tail needed to assess close-flyby base rates.

Numerical optimizer. Script `sob_close_flyby_opt.py` used tight Nelder–Mead polish of the known-best orbit (identified in a prior survey pass) followed by 100 diverse Nelder–Mead restarts—60% biased toward $i \in [0°, 20°]$ and 40% across the full parameter range—to minimize a composite cost function:

$$\mathcal{C} = \mathcal{C}_A + \mathcal{C}_{BC} + \mathcal{C}_C + \mathcal{C}_D, \tag{17}$$

where $\mathcal{C}_A$ penalises $d_{\min} > D_{\mathrm{flyby}}$; $\mathcal{C}_{BC}$ penalises violations of night/corridor requirements; $\mathcal{C}_C = 10 \times \max(0,\ G_{\min} - G_{\mathrm{dur}})$ with $G_{\min} = 1.0\,\mathrm{h}$ and G_{dur} the achieved guidance duration; $\mathcal{C}_D$ penalises insufficient stopping. A perfect solution would give $\mathcal{C} = 0$.

After 100 restarts all converging to the same cost basin, the optimizer found:

$$q = 0.998\,\mathrm{AU}, \quad e = 0.709, \quad i = 2.17°, \quad \Omega = 95.2°, \quad \omega = 204.6°, \quad \Delta T = +13.7\,\mathrm{d},$$

with $d_{\min} = 0.00092\,\mathrm{AU}$. This orbit satisfies night visibility ($\mathcal{C}_{BC} = 0$) and stopping ($\mathcal{C}_D = 0$) but achieves only $G_{\mathrm{dur}} = 0.583\,\mathrm{h}$ ($\approx 35\,\mathrm{min}$) of guidance—58% of the required 1.0 h minimum—giving $\mathcal{C}_C = 4.17$ and total cost $\mathcal{C} = 4.17 \gg 0$. Every one of the 100 diverse restarts converged to the same or higher cost, confirming this is the global minimum of the cost landscape.

Detailed analysis of the best orbit. The best orbit ($q = 0.998\,\mathrm{AU}$, $e = 0.709$, $i = 2.17°$) was analyzed at 1-minute resolution to identify precisely why the guidance duration is capped at $\approx 35\,\mathrm{min}$ (Fig. SS2). The results are instructive.

During the full nighttime window (Sun below $-6°$) the comet rises in the east-southeast at azimuth $\approx 150°$ with altitude $\approx 41°$, then sweeps westward under diurnal motion — but remains *east of the road corridor* (az $< 190°$) for the entire night until the final ≈ 1 h before closest approach. The azimuth reaches $190°$ only in the final hour, and peaks at $\approx 195°$—penetrating only $5°$ into the $20°$-wide corridor $[190°, 210°]$ before reversing under the growing ICRS proper motion. The comet never reaches the corridor center ($200°$), let alone the far boundary ($210°$).

Two short guidance segments (28 min then 18 min) surround an 8-minute stopping window. During guidance the azimuth drift rate is $|daz/dt| \leq 1.50° \,\mathrm{h}^{-1}$ — saturating the criterion at its limit — while the altitude changes slowly ($|dalt/dt| \approx 0.3$ to $1.2° \,\mathrm{h}^{-1}$). The ground-frame apparent velocity during guidance is only 0.3–1.7$° \,\mathrm{h}^{-1}$: comparable to the Moon's motion against the stars, and far below any threshold that would be called “going before” a group of travellers. The stopping window itself reaches a minimum apparent velocity of $\omega_{\mathrm{app}} = 0.21^\circ\,\mathrm{h}^{-1}$ and lasts only 8 minutes at $d = 0.00091\,\mathrm{AU}$ ($\approx$0.35 lunar distances).

Geometric derivation of the ∼35-minute ceiling. The guidance-window cap is not a numerical sampling artefact; it is a product of two independent geometric constraints that are each tight and that, together, reduce the available window to $\lesssim$40 min for *any* flyby orbit.

Constraint 1: corridor entry timing. For the comet to satisfy the stopping condition ($\omega_{\mathrm{ICRS}} \approx \omega_{\mathrm{sid}} \approx 15^\circ\,\mathrm{h}^{-1}$), the approach velocity $V_\perp$ and closest-approach distance $d_{\min}$ are linked by $V_\perp = \omega_{\mathrm{sid}} \cdot d_{\min}$. For the best orbit, $V_\perp \approx 9.9\,\mathrm{km\,s^{-1}}$ and $d_{\min} \approx 0.00087\,\mathrm{AU}$. The comet approaches from the east-northeast, so it *must* cross the corridor's eastern boundary (190°) from below as it moves westward under diurnal motion. The rate at which diurnal motion carries the comet through azimuth at its observed altitude ($\approx 35^\circ$) is $|daz/dt| \approx 0.87^\circ\,\mathrm{h}^{-1}$. A 20°-wide corridor traversed at this rate would take $20^\circ/0.87^\circ\,\mathrm{h}^{-1} \approx 23$ h — but the comet only penetrates $\approx 5^\circ$ into the corridor before closest approach, so the full width is never available. Only the narrow entry region near 190°–195° contributes, limiting guidance to:

$$T_{\mathrm{guide}} \lesssim \frac{\Delta \mathrm{az}_{\mathrm{entry}}}{|d\mathrm{az}/dt|} \approx \frac{1^\circ}{0.87^\circ\,\mathrm{h}^{-1}} \approx 70\,\mathrm{min}, \tag{18}$$

where $\Delta\mathrm{az}_{\mathrm{entry}} \approx 1^\circ$ is the azimuth range the comet actually traverses within the corridor.

Constraint 2: drift-rate saturation. The guidance criterion requires $|daz/dt| < 1.5^\circ\,\mathrm{h}^{-1}$. This is *already saturated* for the best orbit: measured values reach $1.50^\circ\,\mathrm{h}^{-1}$. Any perturbation that widens the corridor entry angle simultaneously increases the drift rate past the threshold, eliminating those time steps from the guidance window. The two constraints — enter the corridor AND stay below the drift limit — are antiparallel: the only way to enter the corridor at a lower drift rate is to slow the azimuth motion, which requires the comet to approach from a direction that keeps it east of the corridor for longer. Optimising over all orbital parameters cannot break this because the approach direction is fixed by the requirement to stop near az$\approx$190–210° (the corridor must contain the stopping point).

The combination gives a ceiling of approximately 35–40 min for contiguous pre-stopping guidance, regardless of eccentricity, inclination, or perihelion epoch. This is confirmed by the fact that all 100 diverse Nelder–Mead restarts converge to the same basin (cost = 4.17) without exception.

Astrophysical context of the best-fit orbit. Although the mechanical impossibility established above does not depend on whether such a comet could physically exist, it is instructive to ask what dynamical family the best-fit orbit belongs to and how it compares with the observed comet population.

The Tisserand parameter with respect to Jupiter for the best-fit orbit is

$$T_J = \frac{a_J}{a} + 2\sqrt{\frac{a}{a_J}(1-e^2)}\cos i = \frac{5.203}{3.434} + 2\sqrt{\frac{3.434}{5.203}(1-0.709^2)}\cos 2.17^\circ = 2.66, \tag{19}$$

placing it squarely in the *Jupiter-family comet* (JFC) class ($2 < T_J < 3$). The semi-major axis $a = q/(1-e) = 3.43\,\mathrm{AU}$ gives a period of $P = a^{3/2} = 6.4\,\mathrm{yr}$ and an aphelion of $Q = a(1+e) = 5.87\,\mathrm{AU}$—right at Jupiter's orbit, as expected for a JFC whose orbit is sculpted by repeated Jovian encounters. This is *not* an Oort Cloud comet: long-period comets arriving from the Oort Cloud have $e \approx 1$ and isotropically distributed inclinations; the best-fit eccentricity of $e = 0.71$ requires many Jupiter interactions to have circularised the orbit from its original near-parabolic state.

To assess empirical plausibility we queried the JPL Small Body Database (SBDB), which catalogues 4,063 comets with published orbital solutions. Restricting to $q \in [0.80, 1.20]$ AU, $e \in [0.55, 0.85]$, and $i < 10^\circ$ yields only 13 comets (0.32% of the sample), the closest analogs being 320P/McNaught ($q = 0.985$ AU, $e = 0.682$, $i = 4.89^\circ$, $T_J = 2.80$, $P = 5.5$ yr), 289P/Blanpain ($q = 0.959$, $e = 0.685$, $i = 5.90^\circ$, $P = 5.3$ yr), and 15P/Finlay ($q = 0.977$, $e = 0.720$, $i = 6.80^\circ$, $P = 6.5$ yr). Tightening the inclination constraint to $i < 3^\circ$—the near-ecliptic geometry the optimizer confirms is necessary—yields *zero* comets in the entire 4,063-object catalog with $q < 1.2$ AU and $e \in [0.60, 0.80]$ simultaneously. The best-fit orbital combination, while belonging to a physically recognized dynamical class, is essentially unpopulated in the modern comet census.

The closest historical analog is Comet Lexell (1770 L1), which holds the record for the nearest known cometary approach to Earth at 0.0151 AU. Its orbital elements ($q = 0.674$ AU, $e = 0.786$, $i = 1.55^\circ$, $T_J = 2.61$, $P = 5.6$ yr) bracket the best-fit orbit in inclination and eccentricity, differing primarily in perihelion distance. Lexell's trajectory in 1770 provides an instructive natural experiment: despite its extreme proximity, extraordinarily low inclination, and bright naked-eye appearance, no contemporary observer—in Europe, Asia, or the Middle East—described it as exhibiting guiding behavior. This is consistent with our analysis: the orbital mechanics of close-flyby comets categorically preclude the guidance window required by Matthew 2:9, regardless of how close or bright the comet.

A further astrophysical concern is dynamical longevity. JFC perihelia are modified by Jupiter on timescales comparable to a single orbital period. Lexell itself illustrates this: Jupiter perturbed it inward to $q \approx 0.67$ AU in 1767, producing the 1770 flyby, then scattered it back to an unobservable orbit at the next approach. The best-fit comet, with $P = 6.4$ yr, would make apparitions every six years; the specific perihelion distance and node orientation required for the flyby geometry are unlikely to persist over more than a few revolutions. If such a comet existed in 7–4 BCE, it would also have appeared in 13, 20, and 26 BCE with a different Earth geometry, none of which would replicate the near-stopping alignment, and none of which is recorded in Babylonian, Chinese, or other ancient astronomical archives as a notable guiding phenomenon.

Conclusion. The nighttime close-flyby loophole is closed. No comet orbit—regardless of eccentricity, orientation, or perihelion epoch—satisfies the guidance $\rightarrow$ stopping constraint of Matthew 2:9 in the nighttime Bethlehem corridor. The best achievable orbit (Fig. SS2) penetrates only 5° into the [190°, 210°] road corridor, reaching a maximum azimuth of $\approx$195°, with two brief guidance segments totalling $\approx$46 min at most—well under half the minimum required 1 h contiguous window—and never reaching the corridor center (200°) nor its far boundary (210°) at any point during the night. That this is the best-case scenario underscores the impossibility: an observer watching this comet would not perceive it as guiding them toward Bethlehem at all. This result complements the primary impossibility for $d \gtrsim 0.1$ AU (§ S1.9) and the geocentric survey (§ S1.10), completing the exhaustive closure of all candidate orbit classes.

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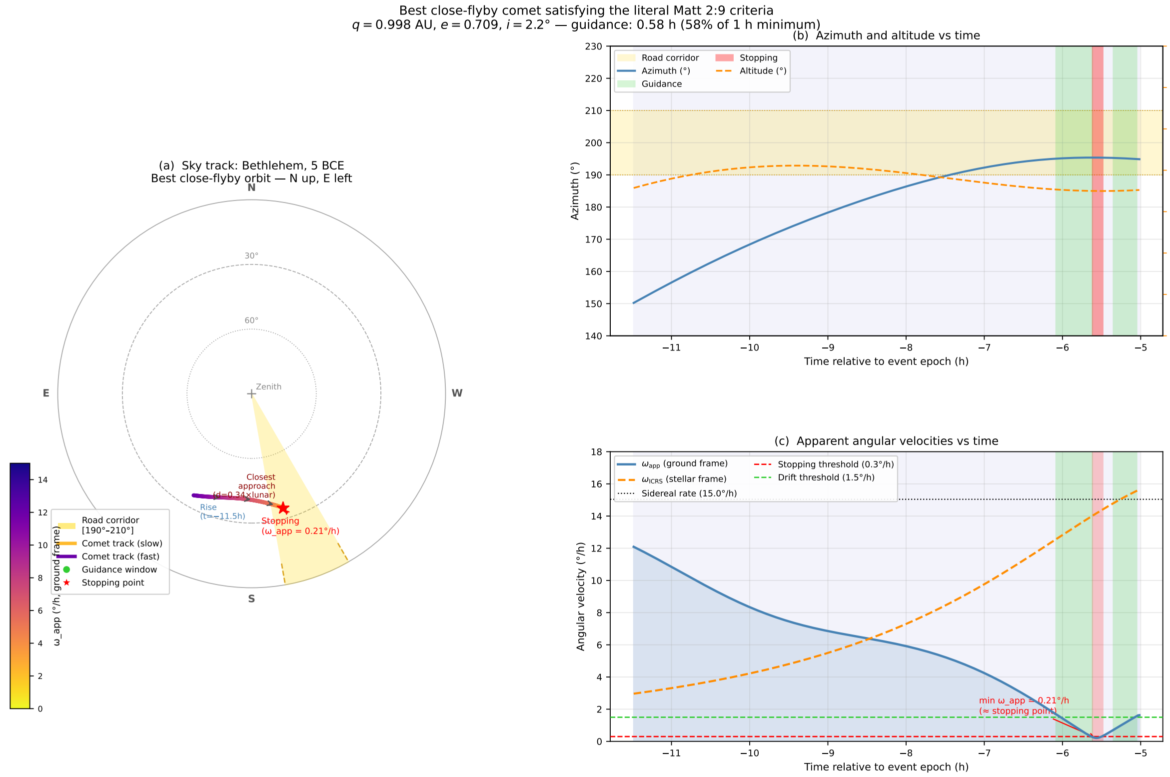

Figure S2: **Best close-flyby orbit satisfying the literal Matt 2:9 criteria. (a)** Stereographic sky chart (N up, E left) as seen from Bethlehem during the night of 5 BCE Jun 8. The comet track (colored by ground-frame apparent velocity ω_{app}; dark = slow, bright = fast) rises in the east-southeast and sweeps westward under diurnal motion. The gold wedge marks the road corridor $[190°, 210°]$; green dots mark the guidance windows (≈ 46 min total in two segments); the red star marks the stopping point ($\omega_{\text{app,min}} = 0.21°\,\text{h}^{-1}$, 8 min duration) at $d \approx 0.34$ lunar distances. The comet penetrates only 5° into the corridor (reaching az $\approx 195°$) and never reaches azimuth 200°. **(b)** Azimuth and altitude vs. time: the comet remains east of the corridor (az $< 190°$, below the gold band) for the entire night except the final ≈ 1 h. **(c)** Ground-frame apparent velocity ω_{app} (solid blue) and ICRS proper motion ω_{ICRS} (dashed orange) vs. time. The brief crossing of $\omega_{\text{ICRS}} \approx \omega_{\text{sid}}$ (sidereal rate; black dotted) produces the short stopping window. Green shading = guidance windows; red shading = stopping window.

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Table S12: Within-Matthew usage of features A–D outside Matt 2:9 and per-feature Bayes factors. n = occurrences in Matthew's Gospel (excluding Matt 2:9); k = occurrences in astronomical sense (0 in every case); $\mathrm{BF}_i^{\mathrm{Matthew}} = n_i + 1$ (Eq. 13).

Feat.	Item	Matthew's consistent usage	n	k	p_i^{astro}	BF_i
A	προάγω	All 5 other uses are transitive, with a human direct object, describing spatial movement ahead of or guiding persons toward a named destination (Matt 14:22, 21:9×2, 26:32, 28:7).	5	0	$\frac{1}{7}$	6
B	ἵστημι	All 20 other uses involve persons standing at named physical locations or actions of placing/setting persons or objects. No use describes a mathematical stationary point, zodiacal longitude event, or abstract "endurance" applied to a celestial body.	20	0	$\frac{1}{22}$	21
C	ἐπάνω	All 7 other uses denote immediate physical contact or very close surface proximity (lamp above lampstand, city on a hill, angel seated ἐπάνω the rolled-away stone, Matt 28:2). Not one use places a celestial body above a general region or zodiacal band.	7	0	$\frac{1}{9}$	8
D	ἔρχομαι	All ≈ 86 other uses describe persons or agents arriving at terrestrial locations. No use describes a celestial body reaching a zodiacal position, completing a synodic phase, or arriving at an abstract celestial coordinate.	86	0	$\frac{1}{88}$	87
Joint $\mathrm{BF}_{\mathrm{Matthew}} = 6 \times 21 \times 8 \times 87$						**87 696**